When one constraint moves, another appears

Climate coping across 24 developing countries, migration policy in Sri Lanka, and a sanitation-finance classic from Cambodia · three peer-reviewed economics-journal articles. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

The common thread is substitution: households respond strongly when a constraint moves, but the response often exposes the next bottleneck. Drought pushes rural households toward bushmeat income, creating conservation and zoonotic-risk externalities. Sri Lanka's migration rule replaces overseas work and foreign remittances with maternal presence and domestic transfers. In Cambodia, financing nearly doubles willingness to pay for latrines but leaves a large installation gap. **Literature trend:** applied development economics is increasingly following the full chain from first-stage behavior to completed welfare outcomes. That shift reveals why a policy can work on its intended margin while relocating costs into implementation, informal coping, or other household resources.

01 CURRENT / CLIMATE SHOCKS AND LIVELIHOOD COPING · STARTS ON PAGE 2 · World Development 208 · 2026 · peer reviewed · CC BY 4.0

Weathering Hard Times with Bushmeat: Rural Household Coping under Rainfall Shocks

How do rural households cope when climate shocks undermine agriculture? The authors convert cross-sectional Poverty and Environment Network surveys into a quarterly panel of nearly 8,000 households in 24 low- and middle-income countries, then match them to gridded moisture stress and local crop calendars. Drought raises bushmeat income—by as much as 39% in an extreme event—and its share of total income, especially during agriculturally critical periods; wet conditions generally reduce both. Household and time fixed effects improve on self-reported shocks, but time-varying local confounding, valuation of informal or illegal income, and the non-random geography of households remain caveats. The paper links climate adaptation to a conservation and public-health externality that standard income measures can miss.

02 CURRENT / MIGRATION AND CHILD HUMAN CAPITAL · STARTS ON PAGE 11 · World Development 200 · 2026 · peer reviewed · Creative Commons

Restricting Mothers' International Migration and Human Capital Investment

What happens when the state keeps mothers of young children from migrating abroad for domestic work? Using Sri Lanka's 2013 Family Background Report rule, repeated household surveys, and a difference-in-differences comparison around the youngest-child age-five cutoff, the authors find less maternal migration, greater maternal presence, fewer child inpatient stays, and lower grade retention among older siblings. Lost foreign remittances are offset by domestic remittances rather than a broad labor-supply response. The design has only limited pre-policy periods and can be affected by age composition, prior exposure, spillovers, and imperfect identification of the migrant. It matters because migration restrictions can improve measured child outcomes while rearranging, rather than eliminating, the household's economic trade-offs.

03 CLASSIC / ENVIRONMENTAL HEALTH AND DEVELOPMENT FINANCE · STARTS ON PAGE 22 · Journal of Environmental Economics and Management 86 · 2017 · peer reviewed classic · CC BY 4.0

Microcredit and Willingness to Pay for Environmental Quality: Evidence from a Randomized-Controlled Trial of Finance for Sanitation in Rural Cambodia

Can credit constraints explain low willingness to pay for environmental health? In 30 rural Cambodian villages, the authors randomize whether households can finance a hygienic latrine and use a Becker-DeGroot-Marschak sales mechanism to trace demand. Financing raises mean willingness to pay from about $30 to $52 and increases the share willing to buy at an unsubsidized price from 25% to 60%, yet only 35–40% of purchased latrines are installed after 18–24 months. Randomly induced purchases by neighbors also reduce a household's own installation. Thirty clusters, loan rejection and cancellation, the demand-elicitation mechanism, and limited ability to separate credit constraints from present bias qualify the results. The enduring lesson is that easing the purchase bottleneck need not complete the health investment.

Editorial note: references and appendices were omitted where possible; no included article contributes more than about 15 text pages, excluding table-heavy pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

World Development 208 (2026) 107522

Contents lists available at ScienceDirect

World Development

journal homepage: www.elsevier.com/locate/worlddev

Regular Research Article

Weathering hard times with bushmeat: rural household coping under rainfall shocks

Prachi Jhamb [a,*], Susana Ferreira [a,c], Martin Reinhardt Nielsen [b]

[a] Department of Agricultural & Applied Economics, University of Georgia, Athens, GA, United States
[b] Department of Food and Resource Economics, University of Copenhagen, Frederiksberg C, Denmark
[c] Institute for Resilient Infrastructure Systems (IRIS), University of Georgia, Athens, GA, United States

ARTICLE INFO

Keywords:
Bushmeat
Hunting Income
Climate shocks
Adaptation
Poverty and Environment Network
Biodiversity Conservation

ABSTRACT

While bushmeat can be a crucial protein source for food insecure households, its consumption and trade raise significant concerns for both human health and wildlife conservation. Here we investigate how rainfall shocks influence rural households' income from and reliance on bushmeat hunting. Drawing on data from nearly 8,000 rural households across 24 low- and middle-income countries, matched to gridded rainfall records and detailed agricultural season calendars, our study distinguishes between the effects of shocks depending on their type, intensity and timing of occurrence within the agricultural cycle. We mitigate endogeneity concerns faced in prior studies relying on self-reported shocks and cross-sectional data, by constructing a quarterly household panel survey integrated with objective, high-resolution data on climatic moisture stress. We find that dry conditions increase both bushmeat income and its share in total household income, particularly during critical agricultural periods. Conversely, wet conditions have the opposite effect, often decreasing hunting due to improved agricultural yields. However, these effects vary depending on the intensity of wet conditions. Additionally, we find heterogeneity in responses across socio-economic strata, as wealthier households tend to have higher bushmeat income during dry periods, likely due to better access to resources. The study increases our understanding of when and by whom climate stressors induce greater wildlife offtake, providing important insight for targeted development and conservation policy.

1. Introduction

Bushmeat, defined as meat from wild animals, primarily mammals, reptiles (such as snakes, crocodiles, lizards, and tortoises), and birds (Fa et al. 2003), serves as an important source of protein for roughly 150 million households in the Global South (Nielsen et al. 2018). Its consumption is associated with a reduced risk of anemia, malnutrition and stunting in children (Fa et al. 2015; Golden et al. 2011), improved household food security (Friant et al. 2020), dietary quality as well as cognitive function (Neumann et al. 2003). In addition to its dietary role, bushmeat serves as a flexible source of income, particularly during low agricultural seasons (Kümpel et al. 2010; Nasi et al. 2008) and in times of crisis, such as those induced by climate shocks (e.g., droughts) (de Merode et al. 2004; van Vliet et al. 2008). Despite these benefits, harvesting and trade of bushmeat also create a policy dilemma. Unsustainable harvesting depletes wildlife populations and threatens biodiversity, while bushmeat trade heightens exposure to zoonotic diseases (Brashares et al. 2011; Pruvot et al. 2019).

Climate change may only amplify these trade-offs. As extreme weather events grow more frequent and severe (IPCC 2023), rural households may increasingly turn to bushmeat hunting to buffer income and consumption shocks (Paumgarten et al. 2018). While existing research (*SM Appendix A*) documents how climate shocks (of different types and intensities) affect household welfare (Carpena, 2019), agricultural productivity (Gornall et al. 2010; Damania et al. 2017), and child health (Hanna & Oliva 2016; Nsabimana & Mensah 2020), their impact on bushmeat hunting (a potential coping mechanism) remains poorly understood. Existing studies on the effect of shocks on bushmeat hunting provide mixed insights. While some suggest bushmeat acts as a drought-induced safety net (Brashares et al. 2011; Mendonça et al. 2016; Paumgarten et al. 2018), others find no evidence to support this claim (Wunder et al. 2014; Nielsen et al. 2015; Nielsen et al. 2017; Nielsen et al. 2018). Similarly, prior research offers mixed evidence on whether bushmeat is primarily a safety net for the poor (Nielsen 2006; Hickey

* Corresponding author.
E-mail addresses: pj40553@uga.edu (P. Jhamb), sferreir@uga.edu (S. Ferreira), mrni@ifro.ku.dk (M.R. Nielsen).

https://doi.org/10.1016/j.worlddev.2026.107522
Accepted 29 July 2026
0305-750X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

et al. 2016; Torres et al. 2021) or an accessible option only for those with sufficient assets (De Merode et al. 2004; Wilkie et al. 2005; Coad et al. 2010; Brashares et al. 2011). Consequently, it remains unclear whether wealth alleviates or intensifies pressure on wildlife.

Critically, previous literature suffers from three limitations: First, prior studies often rely on small-scale case studies or examine wild foods or non-timber forest products broadly (Wunder et al. 2014; Paumgarten et al. 2018), rather than specifically focusing on bushmeat. Second, existing research often aggregates diverse shocks into broad categories (Wunder et al. 2014), potentially obscuring distinct effects of droughts and floods on hunting engagement. Additionally, these studies employ self-reported shock data that can be prone to endogeneity, whereby the households who successfully rely on bushmeat hunting as a coping strategy could also report lower shock intensity. Third, most studies employ cross-sectional research designs, which are prone to omitted variable bias and cannot fully capture households' adaptive responses over time.

Here we investigate how rainfall shocks influence rural households' income from and reliance on hunting, advancing the literature in four ways. First, we leverage the Poverty and Environment Network (PEN) dataset (Angelsen et al. 2011), in which almost 8,000 rural households across 24 countries were interviewed four times within one year, allowing us to build a panel to capture short-term (quarterly) responses to shocks. Second, to measure shocks we employ an objective, high-resolution index of relative water deficit or surplus, the Standardized Precipitation and Evapotranspiration Index (SPEI) (Vicente-Serrano et al. 2010), to mitigate biases inherent in self-reported shock measures. Third, we differentiate between drought and wet conditions, quantify their intensity, and map their occurrence to planting, growing, and harvesting seasons. Finally, we explore heterogeneity by wealth, testing whether adaptive capacity influences households' ability to exploit bushmeat during crises. Specifically, we address the following questions:

1. Do droughts and wet conditions have (different) impacts on households' bushmeat income and reliance?
2. How does the intensity and timing of these shocks within the agricultural calendar affect bushmeat income and reliance?
3. Does households' adaptive capacity, measured by wealth, moderate these effects?

Our results reveal three key findings. First, droughts consistently increase both bushmeat income and reliance, with an extreme drought, increasing hunting income by 39% and reliance on hunting income by 1.8 percentage points. The effects are highest during the growing season, when crop income is most vulnerable. Second, wet conditions reduce bushmeat income and reliance, but only at moderate intensities, suggesting that improved agricultural output lowers hunting. Third, wealthier households exhibit greater capacity to increase hunting income during droughts, reflecting inequalities in adaptive strategies.

The remainder of the paper is organized as follows. Section 2 discusses the data and methods. Section 3 details the empirical strategy. Section 4 discusses the results and robustness checks, and Section 5 concludes.

2. Data and Methods

2.1. Income

Our data originates from the Poverty and Environment Network (PEN) dataset (Angelsen et al. 2011), hosted by the Center for International Forestry Research. Considered the most extensive globally comparable survey on rural households and environmental resource harvesting (Nielsen et al. 2018), the PEN survey covers 8,301 households[1] from 333 villages in 58 sites from 24 countries in Sub-Saharan Africa, Asia and Latin America between 2005 and 2010. The data is geo-referenced at the village level.

One of the key features of the PEN project is its quarterly recording of household income from all sources (environment, agriculture, livestock, wages, business and remittances). Income from these sources encompasses both monetary (cash) income and subsistence income (value of harvested products used directly by the household rather than sold for cash). Income from harvesting bushmeat is part of environmental income which is derived from both forest and non-cultivated environments. Each household in the dataset was surveyed four times over the course of a year, using short recall periods of 1 to 3 months[2] (Wunder et al. 2014). Importantly, survey sites were not selected based on the level of bushmeat hunting or to be nationally representative. Instead, they are located along the full range of forest cover in the countries in which surveys were carried out and to a large extent also along the population density distribution as two important development indicators. The sample is over-weighted by sites in Africa and under-weighted in low forest cover areas with high population densities but is otherwise overall fairly representative of smallholder-dominated tropical and sub-tropical landscapes (Wunder et al. 2014; Nielsen et al. 2018). We therefore do not claim that the PEN survey sites are statistically representative of the countries in which data collection was conducted and interpret our findings as applying to smallholder-dominated tropical and sub-tropical communities with moderate-to-good forest access, the population for whom bushmeat hunting is most relevant. Fig. S1 maps the location of all 333 PEN survey villages across the 24 countries included in this study. A detailed assessment of PEN sample representativeness is provided in *SM Appendix B*.

In this paper, bushmeat income refers to any income, cash or subsistence, obtained from hunting of wild animals[3] for personal consumption, gifting, trading, or medicinal use (Nielsen et al. 2017). Cash and subsistence components are summed prior to applying an Inverse Hyperbolic Sine (IHS) transformation. The recall window for bushmeat income is 1 month. The IHS transformation is applied to accommodate the large share of households reporting zero bushmeat income in any given quarter without dropping those observations. Households reporting bushmeat income, whether cash or subsistence, at any point during the survey period are classified as hunters. Household reliance on bushmeat is expressed as the share of total household income originating from bushmeat, effectively measuring bushmeat's contribution to the overall household economy. All income figures are presented in terms of PPP-adjusted US dollars per adult equivalent units,[4] facilitating cross-household comparisons.

Approximately 39% of surveyed households in the PEN data engage in bushmeat hunting, and bushmeat income contributes around 2% to total household income, of which 89% is consumed in the household (i.

[1] Out of which 7,978 households provided responses to the final survey, which included questions on self-reported shocks over the preceding 12 months.

[2] 1 month for bushmeat income.

[3] We exclude insects, crustaceans, grubs, mollusks, and fish. Our focus is on terrestrial mammals, birds, and reptiles.

[4] To make comparisons between households, we use adult equivalent units (AEU) to standardize the comparisons across different household sizes (such that individuals who are below 15 years of age and adults who are above 65 years of age are assigned a weighting factor of 0.5, whereas all other household members within the age range of 15 to 65 years are assigned a weighting factor of 1). Additionally, for the purpose of comparing income values across different countries, we use purchasing power parity (PPP) rates as specified by the IMF in 2007. These methods are in line with those used by previous studies such as Angelsen et al. (2014); Nielsen et al. (2018) and Noack et al. (2019).

e., subsistence income) (Nielsen et al. 2018). Descriptive statistics (Table S1a, SM) indicate bushmeat hunting practices vary significantly across countries, with reliance on bushmeat income ranging from 10.88% in the sites in Cameroon to 0% (or no reliance) in the sites in India and Ecuador.

2.2. Shocks

We derive our shock variables using the Standardized Precipitation and Evapotranspiration Index (SPEI), introduced by Vicente-Serrano et al. (2010). Because the effect of rainfall on crop yields also depends on the soil's ability to retain water (Harari & Ferrara 2018), SPEI tracks the difference between precipitation and temperature driven evapotranspiration, to better reflect how soil moisture influences plant growth, and more effectively capture wet and dry conditions compared to indices such as the Standardized Precipitation Index (SPI) or the Palmer Drought Severity Index (PDSI) (Vicente-Serrano et al. 2010).

SPEI values can be calculated monthly at various timescales (1 to 48 months) to account for cumulative precipitation deficits or excesses. For instance, the 6-month SPEI is generated by summing up water balance values from the previous six months including the present month. In general, SPEI values calculated over a 3-month, or 6-month timescale are known to effectively capture shocks affecting agriculture, while longer timescales (12 or 24-month) are more appropriate for tracking hydrological shocks (Vicente-Serrano et al. 2010; Homdee et al. 2016; Potop et al. 2014; Tirivarombo et al. 2018; Wang et al. 2021). Accordingly, in this paper we focus on SPEI series calculated at a 6-month timescale (SPEI-6). However, we also verify our baseline results using shorter SPEI at a 3-month timescale (SPEI-3).

Since SPEI values are normalized with a mean of zero and a standard deviation of one, values below minus one (SPEI<-1) (or one standard deviation below the long-term (60-year) mean), represent drought conditions. Conversely, SPEI>1 represents wet conditions. Following convention, we categorize continuous SPEI values into binary variables for drought and wet events and use the thresholds of -2 for droughts and 2 for wet conditions for further classification into severe events (McKee et al. 1993; Paulo et al. 2012; Wang et al. 2014). These thresholds are adopted from the established hydrology and climate science literature and were not selected to maximize statistical significance. We note that SPEI is not a simple precipitation threshold index: because it is constructed from the difference between precipitation and temperature-driven potential evapotranspiration, temperature variation is already embedded within the SPEI signal. This makes SPEI more appropriate for agricultural and livelihood analyses than precipitation-only or temperature-only indices, which each capture only a partial aspect of the same water-balance process. Fig. S2 (SM) displays the distribution of SPEI in our sample.[5] We extracted SPEI values for the PEN village center coordinates and matched these values with household survey data using interview dates. Hence, SPEI values may differ among households within the same village due to variations in interview timing. Table S2 (SM) presents the annual prevalence rates of weather shocks in our sample, as measured by the SPEI. The SPEI values used in this paper are drawn from SPEIbase v2.5, available at http://sac.csic.es/spei, which is based on CRU TS 3.24.01 precipitation and potential evapotranspiration data. The dataset provides monthly SPEI values on a global $0.5^\circ \times 0.5^\circ$ spatial grid covering January 1901 to December 2015.[6]

2.3. Seasons

To assess how seasonal weather fluctuations affect bushmeat hunting income, we adopt a methodological framework similar to Noack et al. (2019). We first align our sample villages with relevant geographical units from Sacks et al. (2010), which stands as the most extensive and current repository of crop planting and harvesting dates available (Challinor et al. 2015). Specifically, Sacks et al. (2010) provides georeferenced information about when each crop is planted and harvested. We use these data to classify each day of the year into one of three seasons: planting, growing, or harvesting, for every village in our sample. We then measure the proportion of household income generated during each season. Finally, this quarterly household income for each season is matched to the timing of drought or wet events. For a more detailed explanation, see *SM Appendix C*.

3. Empirical Strategy

We begin by examining how droughts and wet events affect the key measures of household's engagement in hunting. To do this, we estimate a two-way fixed effects model (Eq. 1):

$$Y_{hjt} = \beta_0 + \beta_1 Drought_{hjt} + \beta_2 Wet_{hjt} + \varepsilon_{hj} + \eta_t + \mu_{hjt} \quad (1)$$

where *h, j,* and *t* index household village and, quarter, respectively. The outcome variable *Y* is either (i) bushmeat income (transformed using the IHS to accommodate zeros), or (ii) reliance on bushmeat (already measured as a percentage). The binary variables *Drought* and *Wet*, indicate whether the household experiences a drought (SPEI < -1) or a wet event (SPEI > 1). Although the survey includes household and village characteristics, such as household size, household head's gender, age, and education, total land ownership, distance to roads/cities, elevation, and number of electrified households, these factors do not vary over time in our dataset. Consequently, they are absorbed by the inclusion of household fixed effects (ε_{hj}). Household fixed effects will also eliminate any unobservable time-invariant differences across households such as fixed geographical attributes (e.g., soil quality, proximity to forests), long-standing cultural attitudes and preferences towards hunting. This ensures that our estimates account for within-household variation, isolating the effect of exposure to shocks on hunting income and reliance.

We also incorporate year fixed effects (η_t) to eliminate factors affecting all households each year (e.g., global macroeconomic conditions). The time index *t* corresponds to the survey quarter in which a household was interviewed. Because each household was tracked for up to four consecutive quarterly interviews within a single calendar year, within-household quarter fixed effects would not be separately identified from household fixed effects. Year fixed effects are therefore preferred: they absorb common shocks affecting all households surveyed in the same calendar year, including global commodity price movements and cross-cutting policy changes, while leaving within-year seasonal variation to be captured by the agricultural season variables in Eq. 2. Standard errors are clustered at the village level (the lowest possible cluster) to address cross-sectional correlation among households within the same geographic unit (villages).

Our main identification assumption is that households are generally unable to predict the exact timing and location of drought or wet events in the short run, making the explanatory variable (weather shock) plausibly exogenous (Musungu et al. 2023). Hence, β_1 and β_2 can be

[5] Please note that SPEI values exceeding the value of 1 are denoted as wet events rather than "floods" due to the complexity of flood development, which cannot be solely attributed to high precipitation levels. Factors such as soil saturation and river discharge also play crucial roles in flood occurrence. However, heavy rainfall is the primary contributor to flood formation (Stephens et al. 2015; Bischiniotis et al. 2018).

[6] Comparing SPEI-measured dry and wet events (Table S2) with those self-reported by the village officials (Table S3) suggests a tendency to overreport weather shocks (Table S4).

interpreted as the causal effect of a drought and wet event on bushmeat income and reliance, conditional on fixed effects.

We acknowledge a potential threat to this identification strategy. Time-varying local confounders, such as temporary wage opportunities, migration, or disease outbreaks, could in principle shift labor toward hunting in the same quarters when droughts occur. However, the most plausible local confounders (reduced farm wages, lower casual labor demand) under incomplete rural labor markets, are themselves likely consequences of the drought. That is, a drought reduces returns to farm labor and simultaneously lowers local agricultural wages, making hunting more attractive. These effects are part of the causal mechanism rather than independent confounders. Shocks from entirely different sources (e.g., road construction, disease outbreaks) are not systematically correlated with SPEI values, which track only water balance conditions, and no specific mechanism has been proposed through which such events would co-occur precisely with drought quarters across 24 countries and 333 villages simultaneously.

3.1. Intensity of Droughts and Wet Events

Next, we differentiate between moderate/severe shocks (SPEI values between -1 and -2 for drought, +1 and +2 for wet) and extreme shocks (SPEI below -2 or above +2). Therefore, we extend Eq. (1) by splitting the binary indicators for Drought and Wet into separate dummies for moderate vs. extreme conditions. The estimation approach allows us to identify whether households' bushmeat income and reliance vary with the severity of drought and wet conditions.

3.2. Seasonal Effects of Weather Shocks

To evaluate whether the impact of a shock depends on the season in which it occurs, we then estimate (Eq. 2):

$$Y_{hjt} = \sum_{i=1}^{3} \alpha_i \, Season_{ihjt} + \sum_{k=1}^{2} \beta_k \, Shock_{kjht} + \sum_{i=1}^{3} \sum_{k=1}^{2} \delta_{ik} \left(Season_{ihjt} \times Shock_{khjt} \right) + \varepsilon_{hj} + \eta_t + \mu_{hjt} \quad (2)$$

where *Season* represents the proportion of a quarter falling into one of three agricultural phases (i = planting, growing, harvesting), and *Shock* indicates the presence of a drought or a wet event. If, for example, a drought strikes during the critical growing season, households may experience crop losses and strategically shift labor toward hunting to offset foregone crop income (*SM Appendix D; Fig. S3*). Because crop income is typically the most important income source (28.7% of total household income) in our dataset (Angelsen et al. 2014), comparing changes in bushmeat income to changes in crop income across agricultural phases can reveal whether bushmeat hunting serves as a compensatory strategy when crops are impacted or as a consistent supplement to agricultural income regardless of crop performance. Accordingly, the outcome variable in Eq. 2 is either i) bushmeat income/reliance, or ii) crop income/reliance. Interaction terms between shock and season (*Season* × *Shock*) reveal whether households' hunting or farming income change based on when a shock occurs in the agricultural cycle. Finally, we note that the specification in Eq. 2 allows for a seasonal reallocation margin: households may naturally have more slack time during the growing season and thus hunt more regardless of weather shocks during that season. This would be captured by the season-in-levels coefficients (the alphas in Eq. 2), which measure baseline hunting patterns absent any shock.

3.3. Heterogeneous Effects by Household Income

Finally, we explore whether the shock-hunting relationship varies by households' capacity to adapt. We measure this capacity using total household income excluding bushmeat income.[7] We rank households into quartiles based on this measure, assigning quartiles within each of the 58 PEN sites so that each household's rank reflects its position relative to others facing the same local ecological, institutional, and market conditions. This within-site assignment avoids the risk that income rank conflates household wealth with between-site differences in agro-ecology, forest access, or market integration. We then estimate:

$$Y_{hjt} = \beta_0 + \beta_1 Shock_{hjt} + \beta_2 Shock_{hjt} \times Income_{hjt} + \varepsilon_{hj} + \eta_t + \mu_{hjt} \quad (3)$$

where Y represents bushmeat income. A statistically significant interaction term (*Shock* × *Income*) would suggest that households with greater incomes, which are arguably better able to cope with or even capitalize on a climate shock, respond differently in terms of bushmeat engagement. The results will help us to better understand the role of economic standing in adaptive behaviors to climate shocks and inform the debate on how wealth relates to reliance on wildlife.

4. Results

4.1. Impact of drought and wet events on bushmeat income and reliance on hunting

Fig. 1 displays our baseline estimates (Eq. (1)) using two different timescales to measure rainfall shocks (SPEI measured over the previous 3 or 6 months). The left panel illustrates results for the full sample, whereas the right panel focuses on the subset of villages in which at least one household hunts ("hunting" sample). Bushmeat income (depicted in blue) is scaled by 100 so that the coefficients can be interpreted as percentage changes, while reliance on bushmeat (shown in red) is measured in percentage points (pp). Confidence intervals are at the 95 percent level. Full regression results for this specification are provided in Table S5 (SM).

The results indicate that droughts significantly increase both bushmeat income and reliance. In the full sample, a drought measured over a 6-month timescale increases bushmeat income by 18% and bushmeat reliance by 0.64 pp.[8] Among households in hunting villages, the impacts are even larger, with a drought increasing bushmeat income by 23% and bushmeat reliance by 0.81 pp. This finding is consistent with evidence from previous studies suggesting that drought-driven food scarcity concentrates large mammals around water sources, making them easier targets for hunters (Lindsey et al. 2011; Fragoso et al. 2022). By contrast, wet conditions have a negative (though often smaller) effect on hunting. A wet event over a 6-month timescale reduces bushmeat income by 15% (full sample) to 17% (hunting-only sample), along with a corresponding decrease in bushmeat reliance by 1.29 pp and 1.52 pp, respectively. Conforming to intuition, results using a shorter timescale of 3 months yield smaller effects, which in the case of wet events are no longer statistically significant at conventional levels.[9]

These findings suggest that households turn to bushmeat hunting as

[7] We exclude bushmeat income from the calculation to avoid endogeneity: including it would mean the grouping variable (income quartile) partially reflects the outcome of interest (bushmeat income). The income measure is quarterly total non-bushmeat income per adult equivalent unit (PPP-adjusted), a time-varying quantity reflecting each household's economic position in the quarter of interview. While a pre shock income measure would be preferable, the PEN dataset does not provide a baseline income variable.

[8] It is important to note that because bushmeat constitutes only about 2% of total household income in the PEN data (Nielsen et al. 2018), even a large absolute increase in bushmeat income translates into a relatively small pp increase in the reliance.

[9] Previous research (Vicente-Serrano et al. 2012 and Kubik & Maurel 2016) also found results using shorter SPEI timescales to be smaller, insignificant or less precise. Wald tests for statistical differences between shock coefficients are presented in Table S6 (SM).

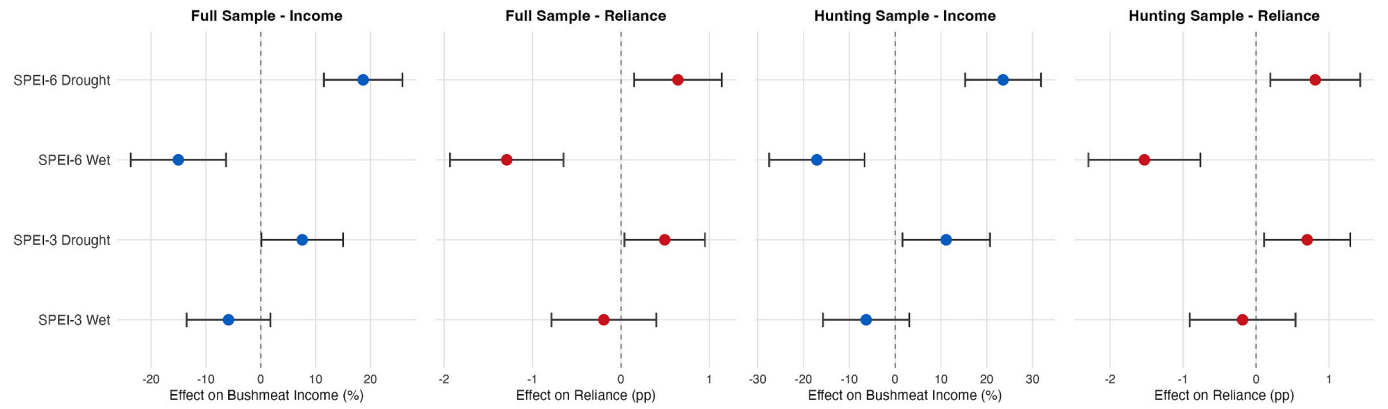


Fig. 1. Effect of Drought and Wet Shocks on Bushmeat Income and Reliance, Full and Hunting Samples. Coefficient plot of estimates from Eq. (1). The two left panels report effects on bushmeat income (blue circles, in %) and reliance on bushmeat income (red circles, in pp) for the full sample. The two right panels report the same outcomes for the hunting sample, defined as households living in villages where at least one household reports bushmeat income in any survey quarter. Income outcome is inverse hyperbolic sine (IHS) transformed, and the reported coefficients are multiplied by 100 so they can be interpreted as percentage changes. Reliance is the share (in %) of bushmeat income in total household income; reported coefficients are in percentage points (pp). All estimated coefficients are at 95% confidence interval. All specifications include household and year fixed effects. Robust standard errors are clustered at the village level. SPEI index is calculated at a 3-month and 6-month time-scale. Drought and Wet dummies are created using the intensity scale of SPEI index, with values below −1 indicating drought conditions and values above + 1 indicating wet conditions.

an adaptive strategy during droughts, presumably to mitigate agricultural losses. In contrast, wet events, which may enhance agricultural productivity, decrease bushmeat income and reliance. Our findings can be explained by a stylized household model (outlined in SM *Appendix D*), where climate-induced shifts in marginal productivity trigger labor reallocation across activities: in the face of drought-induced agricultural income losses, households reallocate labor towards hunting (often a labor-intensive activity that is less sensitive to short run climate shocks). This is in line with the development literature on agricultural household models with incomplete markets (Janvry et al. 1991; Taylor & Adelman 2003; Aragón et al. 2021), where households intensify the use of self-provided inputs such as land or labor to stabilize consumption, when faced with negative income shocks. Hunting, in this context, emerges as a safety net, functioning like other ex-post adaptation strategies, such as sale of livestock, shifting labor to non-farm sources of income generation or migration (Blakeslee et al. 2020; Aragón et al. 2021).

This interpretation finds support in broader evidence (*SM Appendix A*) that while droughts consistently harm agricultural output, the impact of positive rainfall shocks is mixed, potentially benefiting or harming productivity depending on the intensity of rainfall (Damania et al. 2017). For instance, Murken et al. (2024) argue that although extreme wet conditions can lead to waterlogging and crop damage, farmers may still benefit from wet shocks because the relationship between crop yields and climatic conditions is not linear; instead, the optimal yield often occurs with precipitation levels slightly above the average. Prior case studies have also reported that bushmeat hunting tends to rise during the late dry season, when wildlife is easier to locate near limited water sources (Lindsey et al. 2011; Pukazhenthi, 2004).

In addition to our main results based on objective climate data (SPEI indices), we also compare these findings to estimates using self-reported shocks from the PEN survey, presented in Fig. S4 (SM). Interestingly, while the objective SPEI measures show that droughts increase bushmeat income, the self-reported shock data suggest no significant or even opposite effects. This divergence highlights one of our key methodological contributions; relying on objective, spatially consistent climate measures can yield very different insights than self-reported data, which may be affected by endogeneity (*SM Appendix A*).

4.2. Impact of shock intensity on bushmeat income and reliance on hunting

Fig. 2 illustrates the results of the model (Eq. (1)) exploring how households respond to moderate/severe versus extreme droughts and wet events for both the full sample (top panels) and the subset of "hunting" villages (bottom panels). The left panels show changes in bushmeat income, scaled by 100 for percentage interpretation, while the right panels show changes in bushmeat reliance (measured in percentage points), with 95 percent confidence intervals. We only present results for SPEI-6 as it provided more consistent results for both drought and wet conditions in our previous estimation. Full regression results for this specification are provided in Table S7 (bushmeat income) and Table S8 (reliance) (SM).

The results indicate that even relatively less intense droughts prompt households to increase their engagement in hunting. For instance, a moderate/severe drought increases bushmeat income by 17.4% in the full sample and 21.9% in hunting villages, with a corresponding increase in bushmeat reliance by 0.6 pp and 0.7 pp, respectively. As droughts intensify further, the effect becomes notably larger (as confirmed by Wald tests in Table S9, SM). An extreme drought increases bushmeat income by nearly 39% in the full sample and 40.6% in hunting villages, with a corresponding increase in reliance by 1.8 pp and 1.9 pp. These results indicate that households increasingly turn to bushmeat as drought intensity escalates, likely to offset the severe losses in agricultural productivity. In contrast, moderate/severe wet conditions reduce bushmeat income by around 15% (full sample) to 18% (hunting villages) and bushmeat reliance by roughly 1.2 pp and 1.5 pp, respectively. This suggests that since moderate wet episodes replenish soils and enhance crop productivity (Banerjee 2010), they may thereby diminish the need for bushmeat hunting. Extreme wet conditions do not yield statistically significant results.

4.3. Impact of shock timing on bushmeat income versus crop income

Fig. 3 illustrates the effects of shocks on bushmeat income and crop income accounting for the time during the agricultural cycle in which the shock occurs (*Eq. 2)*. We present the results as average partial effects (marginal effects) of drought and wet episodes on bushmeat and crop income. Full regression results for this specification are provided in Table S10 (SM). Estimates of the same specification using bushmeat reliance and crop reliance as dependent variables are presented in Table S11 (SM).

As expected, a drought substantially reduces crop income in the growing season (65% drop). At the same time, a drought increases bushmeat income by around 29% during the growing season (much

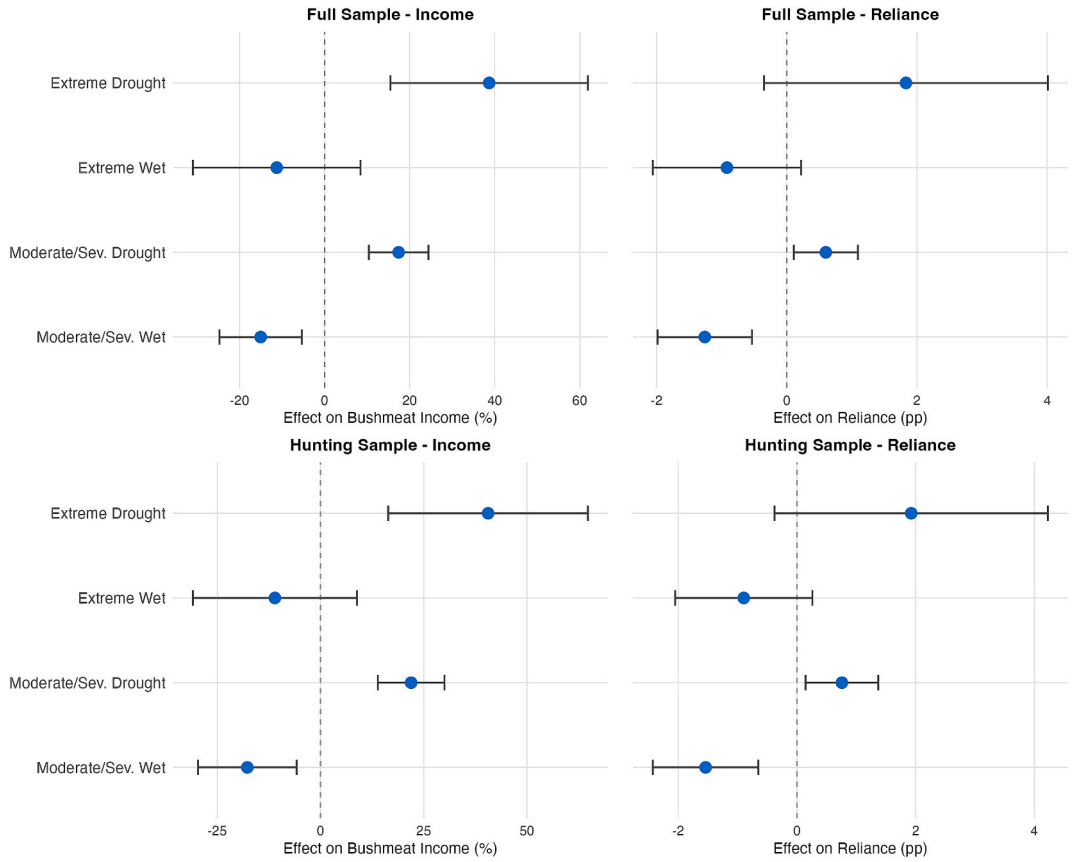


Fig. 2. Effect of Extreme versus Moderate/Severe Shocks (SPEI-6) on Bushmeat Income (%) and Reliance (pp), Full and Hunting Samples. Coefficient plot of estimates from Eq. (1) using a four-category shock specification at the 6-month SPEI timescale. The top row reports the full sample; the bottom row reports the hunting sample. Within each row, the left panel reports effects on bushmeat income and the right panel reports effects on bushmeat reliance. Income outcome is inverse hyperbolic sine (IHS) transformed, and the reported coefficients are multiplied by 100 so they can be interpreted as percentage changes. Reliance is the share (in %) of bushmeat income in total household income; reported coefficients are in percentage points (pp). All estimated coefficients are at 95% confidence interval. All specifications include household and year fixed effects. Robust standard errors are clustered at the village level. Shock dummies are calculated using SPEI index at 6-month timescale, with values below −2, indicating Extreme Drought, values between −1 and −2 indicating Moderate/Severe Drought, values above + 2 indicating Extreme Wet and values between + 1 and + 2 indicating Moderate/Severe Wet conditions.

higher compared to 5% in the planting season and 0.2% in the harvesting season). This supports our hypothesis that when a drought occurs in a period critical for crop growth, households experiencing (and likely anticipating) sharp agricultural losses in the significant future turn to bushmeat hunting to offset those losses. Two pieces of evidence guard against interpreting this growing-season effect as a mechanical seasonal reallocation rather than a genuine shock response. First, the season-in-levels coefficients (α_1, α_2, α_3) in Eq. 2 capture baseline hunting patterns conditional on no shock occurring. Examining these coefficients in Tables S10 and S11 (SM), bushmeat income is higher during the planting season and lower during the harvesting season, the inverse of crop income, and is not elevated during the growing season in the absence of a shock. This confirms that the growing-season spike in hunting is drought-contingent rather than a fixed seasonal pattern driven by slack agricultural labor. Second, a seasonal reallocation story would predict symmetric effects: if households simply exploit slack time in the growing season, they should hunt more regardless of whether conditions are dry or wet. The data show the opposite: bushmeat income falls significantly during wet growing seasons. This directional asymmetry (drought increases hunting, wet conditions decrease it) is inconsistent with a slack-time explanation and instead points to a productivity channel operating through agricultural income losses.

During the planting season, the effects are more pronounced for wet episodes. A wet event during the planting phases can increase crop income through improved soil moisture and increased seedling survival. The same wet event reduces bushmeat income by 41%, suggesting that households allocate less labor to hunting when agriculture is more profitable.

Overall, our results confirm our hypothesis that the timing of climate shocks significantly influences household labor allocation decisions between farming and hunting. These findings align with Noack et al. (2019) who note that livelihood diversification tends to intensify in periods when agricultural income is affected the most and with Costa et al. (2023) who suggest that farmers facing a drought will adapt by adjusting their input mix such as land or labor. Prior case studies have found hunting to intensify in periods following low crop yields (Lindsey et al. 2011) but subside during the height of the agricultural season, when labor shifts back to farming (Brashares et al. 2011; Lindsey et al. 2013).

4.4. *Effects of Income Heterogeneity*

In this subsection we focus exclusively on droughts and on the hunting sample as the foregoing analysis show that the effect of wet conditions are often statistically insignificant and that effects are stronger for households in villages where hunting is prevalent. Fig. 4 (and Table S12, SM) presents the marginal effects of a drought on bushmeat income across the four income quartiles. We find that the magnitude of the drought response is largest for households in the highest income quartile (Q4). Pairwise Wald tests (Table S13, SM) confirm that Q4 is statistically distinguishable from both Q1 and Q3, while differences among Q1, Q2 and Q3 are not statistically significant.

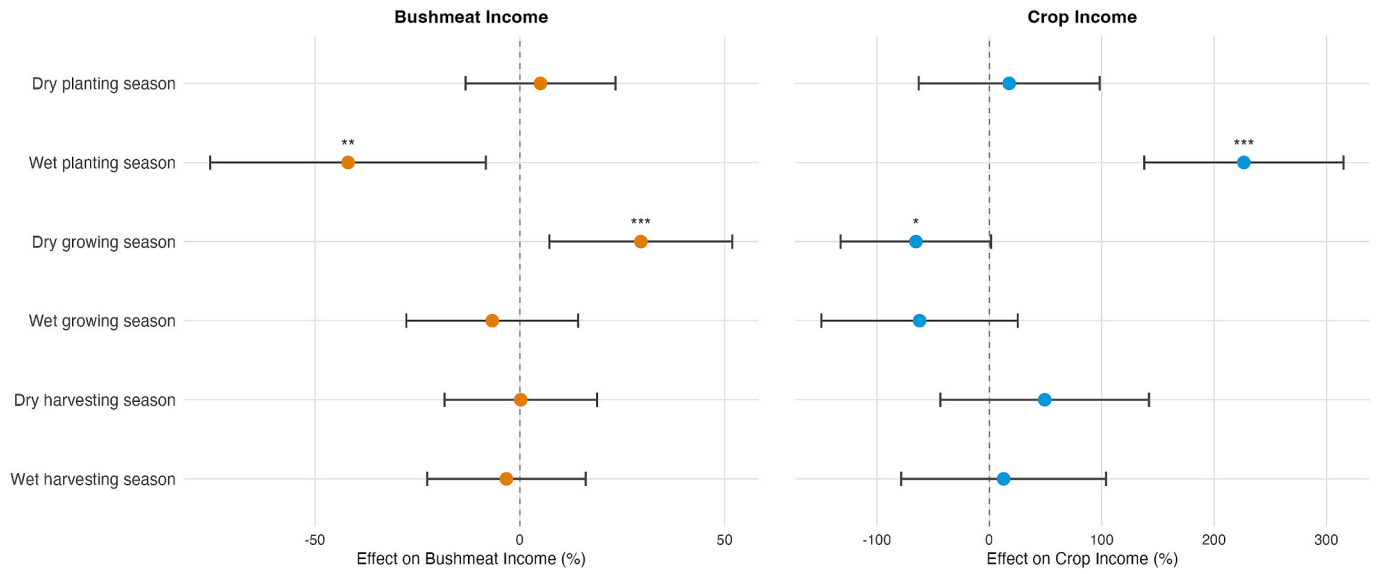


Fig. 3. Marginal Effect of Drought and Wet Shocks on Bushmeat Income (%) and Crop Income (%), by Agricultural Season. Marginal effects from Eq. (2). Coefficients are estimated as the linear combination of the shock effect and the corresponding shock × season interaction (e.g., "Wet planting season" = Wet + Wet × Planting), evaluated relative to the omitted off-season baseline. Both outcomes are IHS-transformed and the displayed coefficients are multiplied by 100 so they can be interpreted as percentage changes (%). All estimated coefficients are at 95% confidence interval. All regressions include household and year fixed effects, and standard errors are clustered at the village level. *p < 0.1; **p < 0.05; ***p < 0.01.

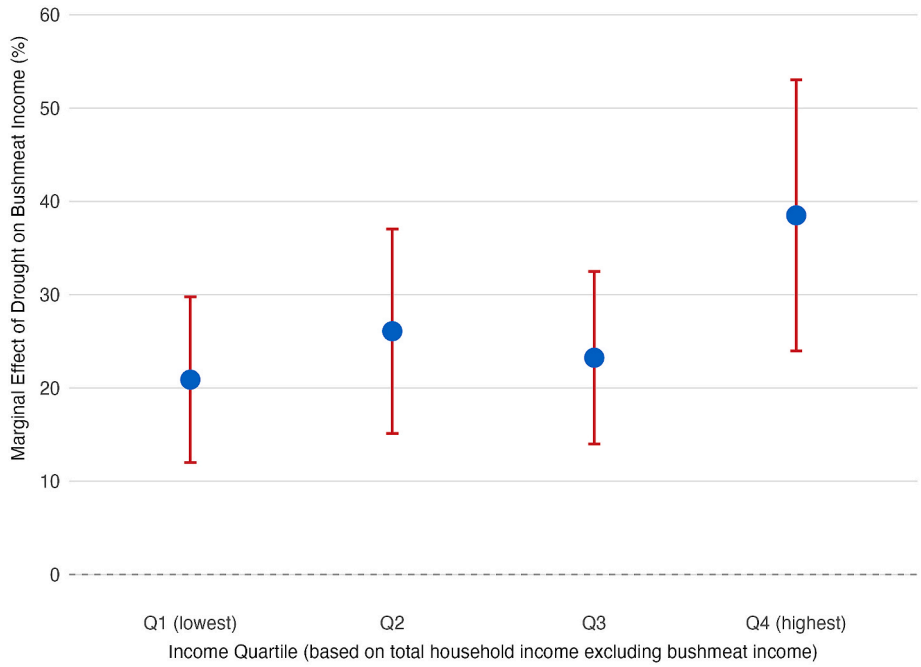


Fig. 4. Marginal Effect of Drought on Bushmeat Income (%), by Income Quartile (Hunting-Prevalent Villages). Income outcome is IHS-transformed and the displayed coefficients are multiplied by 100 so they can be interpreted as percentage changes (%). Quartiles are based on total household income excluding bushmeat income at the site level. Sample is restricted to households in hunting-prevalent villages. All estimated coefficients are at 95% confidence interval. All regressions control for household and year fixed effects, with standard errors clustered at the village level.

These findings can be explained using insights from our exploratory analysis (SM). Although low-income households face droughts more often (Fig. S5, SM) and rely more on bushmeat when they face a drought than high-income households (Fig. S6, SM), they lack the capital or physical resources to scale up hunting when drought hits. High-income households, in contrast, are better positioned to scale up hunting effort: they can invest in specialized hunting equipment (firearms, large-scale traps, and transport), deploy greater household labor on extended hunting trips, and draw on accumulated physical assets that lower the effective cost of intensifying effort when agricultural income falls.

Our findings align with broader literature on forest resource use (Coad, 2008; Arnold & Pérez 2001; Angelsen & Wunder 2003; Heubach et al. 2011) that suggests wealthier households typically derive greater value from forest resources, both in terms of quantity and cash returns from non-timber forest products (*SM Appendix A*). Our results also support previous literature indicating that economic barriers significantly shape access to wildlife resources among different economic groups (Coad et al. 2010; De Merode et al. 2004; Dercon 1998; Ambrose-Oji 2003).

Moreover, previous research suggests that hunting is a risky activity, sometimes involving corruption or bribe payments (Nielsen et al. 2014; Lindsey et al. 2013; Bernard & Stammer 2024). There is evidence that hunters frequently avoid formal penalties by paying bribes when caught (Nielsen et al. 2014), which can be prohibitively expensive for low-income households. Because high-income hunting households can more readily absorb the costs of fines or bribes, they face a comparatively lower risk if caught, whereas low-income hunting households may forgo hunting altogether.

4.5. Robustness Tests

Climate variables tend to be correlated across space, which would cause the standard errors to be biased (Auffhammer et al. 2013). To guard against this, we assess the robustness of our baseline results using Conley's (1999) spatial heteroskedasticity and autocorrelation-consistent (spatial HAC) estimator. Table S14 presents this sensitivity test, confirming that our main findings remain robust.

To ensure our results are not driven solely by changes at the intensive margin, that is, by an increase in bushmeat hunting by households that hunt, we also estimate the probability that a household reports any bushmeat income. Table S15 reports the probability that a household hunts i.e., Pr(bushmeat hunting = 1) using the same two-way fixed effects setup as in our baseline. The results mirror our previous findings. A drought (SPEI-6) significantly increases the likelihood of hunting by 8.7 pp, and a wet event reduces the probability of hunting by 4.5 pp.

To verify that results from heterogeneity by income are not influenced by focusing solely on hunting-prevalent villages, we re-estimate the same specification for the full sample and find that the results remain robust (Table S16).

To assess whether the growing-season drought interaction reflects a genuine agronomic response rather than a spurious seasonal pattern, we implement two additional tests. First, we run a placebo specification in which the two-quarter-ahead (t+2) SPEI value replaces the current-period SPEI in Eq. 2. We use a two-quarter rather than one-quarter lead because the 6-month SPEI accumulation window means that adjacent quarterly SPEI values share approximately three months of overlapping data by construction; the two-quarter lead eliminates this mechanical overlap entirely. Results are reported in Table S17 (SM). Column (1) presents the baseline specification estimated on the placebo sub-sample (N = 7,237). Column (2) replaces current-period drought and wet dummies with their two-quarter-ahead counterparts. Across all eleven coefficients, none is statistically significant at the 10 percent level. The key interaction (Drought × Growing) is effectively zero ($\beta = -0.047$, $p = 0.899$), ruling out anticipation effects and confirming the results are not driven by a spurious seasonal pattern.

Second, we implement a shifted season calendar test. We shift the crop season boundary dates from the Sacks et al. (2010) calendar by ±3 and ±6 months while holding interview dates fixed, and reconstruct planting, growing, and harvesting weights under each shift. Results are reported in Table S18 (SM). The Drought × Growing coefficient is positive and statistically significant only under the correct baseline calendar ($\beta = 0.268$, $p < 0.01$). Under every misaligned specification, displaced by ±3 or ±6 months, the coefficient is statistically indistinguishable from zero, confirming that the drought effect in the growing season is anchored to the agronomic timing of the crop growth stage, inconsistent with a fixed seasonal reallocation.

We also assess robustness to the binary SPEI threshold specification by re-estimating our baseline models with a continuous Drought Index (reversed so that positive values indicate water deficit), entered both linearly and as a quadratic (Tables S19 and S20). The linear specification confirms our main finding: a one standard deviation increase in drought severity increases bushmeat income by 10.7% in the full sample and 12.4% in the hunting sample. The quadratic specification yields an essentially unchanged linear term and a positive, statistically insignificant quadratic term in three of four specifications, indicating the relationship is approximately linear across the observed range of drought severity. Critically, the positive sign on the quadratic is consistent with our binary finding that extreme droughts produce larger effects than moderate droughts. Together, these results confirm that our findings are not an artifact of the specific SPEI threshold of −1.

In the next test, we re-estimate our baseline specification using CRU-derived[10] standardized precipitation anomalies (3-month and 6-month rolling averages) and temperature anomalies, first separately and then jointly (Table S21). Precipitation deficit is positively associated with bushmeat income at the 3-month window but is not significant at 6 months. In the joint specification including both precipitation and temperature anomalies, effects are only jointly significant at the 3-month accumulation, and neither raw input individually replicates the precision of SPEI. This is expected: SPEI combines precipitation and temperature-driven evapotranspiration into a single water-balance signal, whereas the two raw meteorological inputs each capture only one dimension of that process. These patterns confirm that our main findings are directionally robust to alternative climate metrics, and that SPEI outperforms its components as a predictor of bushmeat hunting responses.

To verify that weather shocks do not drive selective survey attrition, we construct a balanced panel covering all household and village-quarter combinations and estimate a linear probability model, regressing a binary interview indicator on drought and wet dummies. Neither drought nor wet conditions are statistically associated with interview participation, ruling out weather-driven selective attrition (Table S22).

5. Conclusion

This paper combines high-resolution gridded climate data with quarterly income records from the Poverty and Environment Network across 24 countries (surveyed between 2005 and 2010) to provide causal evidence on how rainfall shocks affect rural households' engagement with bushmeat hunting in smallholder-dominated forest adjacent tropical and sub-tropical landscapes.

First, our results consistently show that droughts have a significant and positive effect on income from and reliance on bushmeat. A moderate drought can increase bushmeat income by 18%, and up to 39% under extreme drought conditions. By contrast, a moderate wet event can reduce bushmeat income by 15%, likely because improved crop yields reduce the need for alternative income, although extreme wet conditions do not produce a significant effect. Second, by exploiting data on the seasonal calendar, we show that droughts during the growing season lead to the largest increase in bushmeat income (roughly 29% increase), highlighting a strategic reallocation of labor when crop incomes are most at risk. Finally, heterogeneity analysis suggests that, when faced with a drought, wealthier households are better positioned to extract higher bushmeat income, even though poorer households rely more heavily on bushmeat as a proportion of their income. In other words, the coping strategy of bushmeat hunting is available to both, but the wealthier can exploit it more effectively, due to better equipment, access, and perhaps greater ability to handle the costs or risks of hunting. Nonetheless, our findings confirm that bushmeat remains a critical fallback option during droughts, corroborating prior evidence that hunting serves as a safety net during periods when agricultural income or other sources of income are disrupted (Jambiya et al. 2007; Ordaz-Németh et al. 2017; Schulte-Herbrüggen et al. 2013; Enns et al. 2023). Importantly, these estimates capture short run, within-household coping responses to transient weather shocks; they should not be extrapolated to represent long run structural changes in hunting pressure.

Taken together, our paper contributes to the literature in two ways. Methodologically, we show the value of integrating objective measures

[10] Constructed directly from the CRU TS3.21 gridded meteorological dataset, standardized against the full 1901–2012 baseline.

of climate shocks with household income surveys, to overcome the biases of self-reported shock data. Empirically, we disentangle the effect of droughts versus wet events, which prior studies reliant on coarse shock measures have obscured. Our findings carry implications for policymakers addressing rural resilience, conservation, and zoonotic disease risks. Limiting access to forest resources through exclusionary conservation could jeopardize the livelihoods of small-scale landholders in tropical and sub-tropical communities, particularly given the role of bushmeat as a "climatic shock-absorber." By identifying when and for whom bushmeat serves as a crisis buffer, our paper suggests how interventions such as weather index insurance (Barnett & Mahul 2007; Janzen & Carter 2019) should be targeted to reduce hunting pressure while safeguarding vulnerable households.

We acknowledge one important limitation concerning external validity. The PEN survey sites were not designed to be nationally representative; PEN communities do, however, tend to have higher forest cover and greater proximity to protected areas than random tropical locations. Given these characteristics, we do not extrapolate our findings to low forest-cover, high-density peri-urban settings, but note that our target population, forest adjacent rural households, represents the precise context in which the bushmeat–climate nexus is policy-relevant.

CRediT authorship contribution statement

Prachi Jhamb: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Susana Ferreira:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Martin Reinhardt Nielsen:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Acknowledgements

This work was supported by National Institutes of Health, NIH R01AI156866 'Spillover of Ebola and other filoviruses at ecological boundaries'. The funder played no role in the study design, data collection or analysis, decision to publish, or preparation of the manuscript.

The authors sincerely thank 33 PEN partners for data collection, and the Center for International Forestry Research (CIFOR), especially Arild Angelsen, for providing access to the geolocation data from the PEN surveys. We are also deeply grateful to Patrick Stephens, Mekala Sundaram, Shanjukta Nath and Nicole Gottdenker for their insightful comments and valuable support that significantly improved the quality of this paper. Special appreciation is extended to Fredrick Noack for his generous assistance and guidance in replicating the findings detailed in his earlier publication (Journal of the Association of Environmental and Resource Economists, 2019, 6(4), 823-852). Finally, we appreciate the helpful feedback and suggestions from seminar participants at ONSWEET@AERE, particularly Ariel Ortiz Bobea.

Appendices. Supplementary materials

Supplementary data to this article can be found online at https://doi.org/10.1016/j.worlddev.2026.107522.

Data availability

The authors do not have permission to share data.

References

Ambrose-Oji, B. (2003). The contribution of NTFPs to the livelihoods of the 'forest poor': Evidence from the tropical forest zone of south-west Cameroon. *International forestry review, 5*(2), 106–117. https://doi.org/10.1505/IFOR.5.2.106.17420

Angelsen, A., & Wunder, S. (2003). Exploring the forest-poverty link. *CIFOR occasional paper, 40*, 1–20. https://doi.org/10.17528/cifor/001211

Angelsen, A., Larsen, H.O., Lund, J.F., Smith-Hall, C., Wunder, S., (eds.) 2011 Measuring livelihoods and environmental dependence: Methods for research and fieldwork. Center for International Forestry Research (CIFOR), Bogor, Indonesia. DOI: http://dx.doi.org/10.17528/cifor/003341.

Angelsen, A., Jagger, P., Babigumira, R., Belcher, B., Hogarth, N. J., Bauch, S., & Wunder, S. (2014). Environmental income and rural livelihoods: A global-comparative analysis. *World development, 64*, S12–S28. https://doi.org/10.1016/j.worlddev.2014.03.006

Aragón, F. M., Oteiza, F., & Rud, J. P. (2021). Climate change and agriculture: Subsistence farmers' response to extreme heat. *American Economic Journal: Economic Policy, 13*(1), 1–35. https://doi.org/10.1257/pol.20190316

Arnold, J. M., & Pérez, M. R. (2001). Can non-timber forest products match tropical forest conservation and development objectives? *Ecological economics, 39*(3), 437–447. https://doi.org/10.1016/S0921-8009(01)00236-1

Auffhammer, M., Hsiang, S. M., Schlenker, W., & Sobel, A. (2013). Using weather data and climate model output in economic analyses of climate change. *Review of Environmental Economics and Policy*. https://doi.org/10.1093/reep/ret016

Banerjee, L. (2010). Effects of flood on agricultural productivity in Bangladesh. *Oxford Development Studies, 38*(3), 339–356. https://doi.org/10.1080/13600818.2010.505681

Barnett, B. J., & Mahul, O. (2007). Weather index insurance for agriculture and rural areas in lower-income countries. *American Journal of Agricultural Economics, 89*(5), 1241–1247. https://doi.org/10.1111/j.1467-8276.2007.01091.x

Bernard, A., & Stammer, E. G. (2024). *Balancing Tradition and Conservation.*

Bischiniotis, K., Van Den Hurk, B., Jongman, B., Coughlan de Perez, E., Veldkamp, T., De Moel, H., & Aerts, J. (2018). The influence of antecedent conditions on flood risk in sub-Saharan Africa. *Natural Hazards and Earth System Sciences, 18*(1), 271–285. https://doi.org/10.5194/nhess-18-271-2018

Blakeslee, D., Fishman, R., & Srinivasan, V. (2020). Way down in the hole: Adaptation to long-term water loss in rural India. *American Economic Review, 110*(1), 200–224. https://pubs.aeaweb.org/doi/pdfplus/10.1257/aer.20180976.

Brashares, J. S., Golden, C. D., Weinbaum, K. Z., Barrett, C. B., & Okello, G. V. (2011). Economic and geographic drivers of wildlife consumption in rural Africa. *Proceedings of the National Academy of Sciences, 108*(34), 13931–13936. https://doi.org/10.1073/pnas.1011526108

Carpena, F. (2019). How do droughts impact household food consumption and nutritional intake? A study of rural India. *World Development, 122*, 349–369. https://doi.org/10.1016/j.worlddev.2019.06.005

Coad, L. M. (2008). *Bushmeat hunting in Gabon: socio-economics and hunter behaviour* (Doctoral dissertation, University of Cambridge). https://www.repository.cam.ac.uk/handle/1810/252091.

Coad, L., Abernethy, K., Balmford, A., Manica, A., Airey, L., & Milner-gulland, e. j.. (2010). Distribution and use of income from bushmeat in a rural village, central Gabon. *Conservation biology, 24*(6), 1510–1518. https://doi.org/10.1111/j.1523-1739.2010.01525.x

Coad, L., Willis, J., Maisels, F., Funk, S. M., Doughty, H., Fa, J., & Nasi, R. (2021). September). Impacts of taking, trade and consumption of terrestrial migratory species for wild meat. *Convention on Migratory Species.* https://hdl.handle.net/10568/116180.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of econometrics, 92*(1), 1–45. https://doi.org/10.1016/S0304-4076(98)00084-0

Costa, L., Sant'Anna, A. A., & Young, C. E. F. (2023). Barren lives: Drought shocks and agricultural vulnerability in the Brazilian Semi-Arid. *Environment and Development Economics, 28*(6), 603–623. https://doi.org/10.1017/S1355770X21000176

Challinor, A. J., Parkes, B., & Ramirez-Villegas, J. (2015). Crop yield response to climate change varies with cropping intensity. *Global change biology, 21*(4), 1679–1688. https://doi.org/10.1111/gcb.12808

Damania, R., Desbureaux, S., Hyland, M., Islam, A., Rodella, A. S., Russ, J., & Zaveri, E. (2017). *Uncharted waters: The new economics of water scarcity and variability*. *World Bank Publications*. https://doi.org/10.1596/978-1-4648-1179-1

De Merode, E., Homewood, K., & Cowlishaw, G. (2004). The value of bushmeat and other wild foods to rural households living in extreme poverty in Democratic Republic of Congo. *Biological conservation, 118*(5), 573–581. https://doi.org/10.1016/j.biocon.2003.10.005

Dercon, S. (1998). Wealth, risk and activity choice: Cattle in Western Tanzania. *Journal of Development Economics, 55*(1), 1–42. https://doi.org/10.1016/S0304-3878(97)00054-0

Enns, C., van Vliet, N., Mbane, J., Muhindo, J., Nyumu, J., Bersaglio, B., & Nasi, R. (2023). Vulnerability and coping strategies within wild meat trade networks during the COVID-19 pandemic. *World Development, 170*, Article 106310. https://doi.org/10.1016/j.worlddev.2023.106310

Fa, J. E., Currie, D., & Meeuwig, J. (2003). Bushmeat and food security in the Congo Basin: Linkages between wildlife and people's future. *Environmental conservation, 30*(1), 71–78. https://doi.org/10.1017/S0376892903000067

Fa, J. E., Olivero, J., Real, R., Farfán, M. A., Márquez, A. L., Vargas, J. M., & Nasi, R. (2015). Disentangling the relative effects of bushmeat availability on human nutrition in central Africa. *Scientific reports, 5*(1), 8168. https://doi.org/10.1038/srep08168

World Development 200 (2026) 107284

Contents lists available at ScienceDirect

World Development

journal homepage: www.elsevier.com/locate/worlddev

Regular research article

Restricting mothers' international migration and human capital investment

Takuya Hasebe [a,*], Yuma Noritomo [b], Bilesha Weeraratne [c]

[a] Sophia Institute for Human Security and Faculty of Liberal Arts, Sophia University, Tokyo, Japan
[b] Charles H. Dyson School of Applied Economics and Management, Cornell University, Ithaca, NY, USA
[c] Institute of Policy Studies of Sri Lanka Colombo Sri Lanka

HIGHLIGHTS

- We analyze the effect of a unique policy in Sri Lanka that restricts mothers with young children from migrating internationally for employment on human capital investment.
- The policy reduced international migration and increased mothers' presence at home.
- It led to improved child health (fewer inpatient stays) and positive spillovers on older siblings' education (reduced grade retention).
- The drop in foreign remittances was offset by an increase in domestic remittances.

ARTICLE INFO

Keywords:
Human capital
Health
Education
Remittances
Sri Lanka

ABSTRACT

International migration offers significant economic opportunities for developing countries, but it can also separate parents from their children, potentially harming child development. This paper examines the effects of restricting mothers' international migration on left-behind children, leveraging a unique Sri Lankan policy that restricted mothers with children under age five from migrating abroad for employment. Using a difference-in-differences approach, the results reveal the following: First, the policy reduces international migration, increasing mothers' presence at home. Second, policy exposure leads to better healthcare outcomes, including a significant reduction in inpatient stays, particularly treatment for illnesses. This improvement appears to result from increased child-care and monitoring by mothers. Although the policy decreases remittances from abroad, this reduction is offset by an increase in domestic remittances. Furthermore, we find evidence of positive spillovers on non-targeted children with younger, policy-targeted siblings, as indicated by reduced grade retention. These findings highlight the trade-offs between a mother's presence and the economic opportunities associated with international migration in shaping human capital development.

1. Introduction

International migration plays an important role in developing countries. Remittances remain a crucial source of external finance for low- and middle-income countries. Officially recorded remittance flows to these countries reached an estimated $656 billion in 2023 (World Bank, 2024). Remittances provide people in low-income countries with higher incomes and greater economic opportunities, and they are linked to improved child outcomes, including better educational outcomes and reduced child labor in sending communities (Alcaraz et al., 2012; Edwards and Ureta, 2003). However, such migration opportunities, especially for mothers, may also adversely affect the children left behind due to reduced interaction or monitoring by their mothers (Cortes, 2015; Meng and Yamauchi, 2017). The overall effect of parental migration—particularly mothers'—on children remains unclear, as it entails a trade-off between the income gains from international migration and the loss of maternal presence at home.

This paper examines the effects of restricting mothers' international migration by leveraging a unique policy in Sri Lanka. Historically, Sri Lanka has sent a large number of female migrants internationally, mostly as domestic workers in the Middle East. This migration has been an important income source for both households and the country. Recently, however, there is a growing concern around well-being of children left behind, where a mother's absence leads to child neglect and various adverse consequences (Abeyasekera and Jayasundere, 2015). This concern led to the policy of the Family Background Report (FBR, hereafter)

* Corresponding author.
Email addresses: thasebe@sophia.ac.jp (T. Hasebe), yn266@cornell.edu (Y. Noritomo), Bilesha@ips.lk (B. Weeraratne).

https://doi.org/10.1016/j.worlddev.2025.107284

Available online 13 December 2025
0305-750X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

in 2013, such that mothers who have children aged below 5 are not allowed to migrate internationally for employment as domestic workers. This provides an ideal setting to test the trade-offs associated with maternal migration on child development.

Our empirical analysis relies on a difference-in-differences (DID) approach to identify and estimate the impacts of restricting mothers' international migration on child development and other household-level outcomes which potentially mediate the impacts, using repeated cross-sectional data from the nationally representative Sri Lanka Household Income and Expenditure Survey. The DID approach employs two dimensions of comparison. First, we compare households with the youngest children above and below age 5, as the restriction applies to mothers with children under 5. Second, we compare outcomes from the years before (the 2009/2010 and 2012/2013 waves) and after the policy (the 2016 wave). We restrict our analytical sample to households whose youngest child is aged 2–10, to mitigate concerns related to fertility responses and differences associated with secondary school enrollment. The DID approach effectively isolates the effects of interest, assuming the parallel trends assumption holds—that households with the youngest child under 5 and those with the youngest child over 5 would have followed similar trends absent the policy.

The results indicate that the policy improved child health by successfully discouraging mothers from migrating internationally and encouraging them to stay at home. First, we find that policy exposure leads to a 1.5 percentage point decrease in the likelihood of any household member migrating abroad, relative to the control mean of 7.6 %. This decrease seems to be driven by the reduction in mothers' migration abroad. Furthermore, this is accompanied by a 1.2 percentage point increase in the likelihood of a mother's presence at home, relative to the control group of 97 %.

Second, the policy induces better child health outcomes. We examine each child's healthcare utilization, intending to capture underlying health conditions, alongside an analysis of chronic diseases. While we do not find statistically significant effects on outpatient visits for treatment for illness or for check-ups, we find that the policy significantly decreases the likelihood of any inpatient stay by 1.1 percentage points, relative to the control mean of 7.2 %, representing a 15 % decrease. Inpatient stays specifically for treatment for illness decrease significantly by 0.8 percentage points from the control mean of 5.7 %, a 14 % reduction. We do not observe significant effects on chronic disease, which may be too early to diagnose in the children included in our study as it may take time for any potential effects to manifest. These findings hold up in a battery of robustness checks, such as prior treatment exposure, different subsamples, and a falsification test using a pseudo cutoff age.

The effects on mothers' presence and improved child health are closely linked, with the increase in child health primarily driven by the mother's greater presence at home. A mother's presence may reduce the likelihood of illness by enabling greater investment in her child's health capital. The policy resulted in mothers staying at home to care for their children, which, in turn, contributed to improved child health outcomes.

We then examine another potential channel: income effect. It is unlikely to be the main driver of the observed effects. Specifically, a negative income effect could arise if restricting mothers' international migration reduces household income due to decreased international remittances. Our results show that while policy exposure significantly reduces remittances from abroad, domestic remittances increase correspondingly. Therefore, there is no statistically significant impact on total remittances (international and domestic combined) or overall household income. Given the absence of changes in household income and Sri Lanka's free universal healthcare system, the observed decrease in inpatient stays is more plausibly attributed to improved health status rather than financial constraints limiting healthcare access.

We next extend our analysis to examine whether the observed effects of improved child health are primarily driven by the policy target—children under 5 years old. It is important to note that the children who were treated consist of not only these target children but also their older siblings. In other words, our estimated effects capture both the direct impact on the target population and the indirect effects on their siblings. Our analysis confirms that the main results are driven by the policy-targeted children, showing significant direct but no indirect effects.

We also examine spillover effects on non-targeted children's educational outcomes, defined only for school-aged children.[1] The results suggest positive spillover effects. While policy exposure does not significantly increase school attendance (control mean: 98.4 %), it is associated with a statistically significant reduction in grade retention. This finding also highlights the importance of current mother–child interactions.

The contribution of this paper is to evaluate the net effects of restricting mothers' international migration and examine the potentially conflicting mechanisms driving these effects. Two features of the Sri Lankan policy make it especially unique and interesting. First, it is restrictive rather than expansionary, allowing us to test for asymmetries relative to the migration-promoting interventions that dominate existing evidence.[2] Second, it is gender-targeted, applying only to mothers. Because much of the literature comes from male-dominant migration contexts,[3] examining gender differences in impacts is essential. Our results complement this work by showing that the restriction-induced increase in maternal presence positively affects human-capital investment in children.

In the remainder of the paper, Section 2 lays out the conceptual framework and reviews key related literature. Section 3 describes the Sri Lankan background and the policy. Section 4 and Section 5 explain the data and econometric strategy, respectively. Section 6 presents the results, and Section 7 concludes.

2. Related literature and conceptual framework

The policy restricting mothers' international migration may affect children through two pathways: (i) a positive effect via increased maternal presence, interaction, and monitoring, and (ii) a negative effect via reduced international remittances.

Mothers' presence may be particularly important in the context of migration. First, existing evidence raises concerns that migration-induced parental separation harms left-behind children. Second, the human capital literature emphasizes that maternal time input is critical for children's human capital development. For example, in the context of China's rural-urban migration, being left behind by both parents significantly impairs children's cognitive development (Zhang et al., 2014) and health (Li et al., 2015), whereas the effects are much smaller and insignificant when only one parent is absent. Meng and Yamauchi (2017) show that maternal urban migration adversely affects the health and educational outcomes of rural children in China. Similarly, Bai et al. (2022) find negative effects of maternal migration on the critical cognitive development of left-behind children in China. Outside China, Botezat and Pfeiffer (2020) link parental migration to serious health problems and depression in Romania. In a comparative study, Viet Nguyen (2016) finds that parental migration negatively affects child health outcomes in India, Peru, and Vietnam, but has no significant effect in Ethiopia. Cortes (2015) examines the gendered impacts of parental migration on education in the Philippines—a country with a high share of female migrants, similar to Sri Lanka—and shows that a mother's absence has a more pronounced detrimental effect than a father's. Unlike previous studies,

[1] In Sri Lanka, the school starting age is 5 years, coinciding with the migration policy cutoff.

[2] For example, several studies have used randomly assigned visa lotteries or cash transfers to identify the causal effects of migration (Bryan et al., 2014; Gibson et al., 2011, 2018; McKenzie and Yang, 2012; Mobarak et al., 2023).

[3] For example, empirical studies have shown that male migration reduces the labor supply for market work among left-behind females in countries such as Mexico (Amuedo-Dorantes and Pozo, 2006), Egypt (Binzel and Assaad, 2011), and Albania (Mendola and Carletto, 2012).

we assess mothers' presence—rather than separation—by leveraging a policy that restricted their international migration.

In the human capital literature, the importance of maternal time investment for children's human capital accumulation has been well recognized (Francesconi and Heckman, 2016). Early childhood—particularly the preschool period—is a critical window for development with lasting consequences (Almond et al., 2018; Luby et al., 2016). For example, Brooks–Gunn et al. (2002) and Waldfogel et al. (2002) show that early maternal employment, which reduces maternal care time, has been associated with lower cognitive development. In the migration context, Bai et al. (2022) find that maternal migration is detrimental to preschool-aged children's cognitive development. The FBR policy primarily targets children under age 5 by restricting their mothers from migrating, with the explicit aim of ensuring maternal presence during this critical stage.

On the other hand, a complementary strand of research documents the importance of remittances for left-behind families—particularly children. Mobarak et al. (2023) examine migration from Bangladesh to Malaysia and find that increased remittances raise the living standards of migrants' families. Carletto et al. (2011) report higher height-for-age z-scores and lower stunting among children in Guatemalan households with a migrant in the United States. Several studies find positive effects of remittances on schooling outcomes (e.g., Antman, 2012; Edwards and Ureta, 2003). In the Philippines, Yang (2008) shows that positive remittance shocks increase human-capital investment, including school attendance and education expenditures, whereas in Mexico, Alcaraz et al. (2012) show that negative remittance shocks raise child labor and reduce school attendance.

Because the two policy channels—greater maternal presence and reduced international remittances—operate in opposite directions, the net effect on children is ultimately an empirical question.

In addition, the policy impacts on maternal presence and international remittances may be attenuated if households cope with the restriction by reallocating labor or adjusting remittances. Migration decisions are made jointly within the family (Stark and Bloom, 1985). In response to a ban on mothers' international migration, households may reallocate labor to offset lost earnings–fathers or other members might migrate abroad or domestically to send remittances back instead of mothers, or increase labor supply and income in local labor markets. Mothers themselves may pursue domestic migration to increase domestic remittances, since the policy restricts only international moves. Households may also draw on existing networks for support–for example, receiving remittances from relatives in migration hubs beyond the immediate family (Stark and Lucas, 1988). Recent evidence suggests that domestic migration networks and intra-household labor diversification can buffer income shocks through remittances (Batista and Vicente, 2023; Bettin et al., 2024). Understanding both mothers' responses and broader household adjustments is therefore essential to assess the overall policy's impact on children. We explore this channel in the mechanism analysis.

3. Background

Sri Lanka has sent more than 200,000 migrants every year since 2002, and the number reached a peak of 300,000 in 2014 (SLBFE, 2018). This scale of international labor migration is notable, considering the total labor force was approximately 8 million in 2014. This migrant labor contributes to the economic development of the nation by sending remittances, which are one of the nation's main sources of foreign revenue earnings. Remittances amounted to US$ 6.4 billion and accounted for 8.3 % of GDP in 2013 (World Bank, 2015). In the same year 2013, approximately 40 % of the migrants were female, and more than 80 % of them worked as domestic workers, with the Middle Eastern countries as popular destinations. Most labor migrants from Sri Lanka are recruited through registered foreign employment agencies under typically two-year contracts, making migration temporary in nature. Re-migration often occurs when migrants fail to meet their intended goals—such as savings or investment targets—or when returnees encounter difficulties in reintegrating into society.

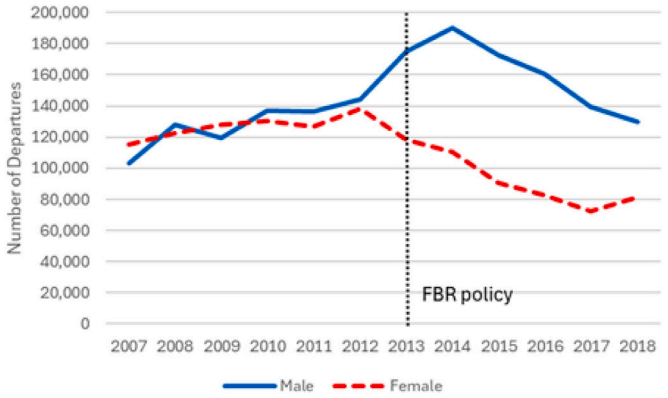


Fig. 1. Trends in departures for foreign employment from 2007 to 2018 in Sri Lanka. Notes: Source: Sri Lanka Bureau of Foreign Employment (2018). "FBR policy" refers to the introduction of the FBR, which began in 2013. The y-axis represents the number of departures, defined as the number of individuals departing for foreign employment who are registered with the Sri Lanka Bureau of Foreign Employment.

Although the migrant labor has brought the benefits to the country, it has also imposed costs on household members left, particularly children. To protect the welfare of children separated from their mothers, the Sri Lankan government took a policy action by issuing Circular 13/2013 in June 2013. The Circular requires female domestic workers to fill out a Family Background Report (FBR) as a pre-departure requirement, which came into effect on 15th July 2013. It, in principle, restricts female domestic workers with children under the age of 5 from migrating internationally for employment. Initially, the Circular covered females who sought employment in domestic worker jobs abroad, but in August 2015, its coverage was expanded to all female employment abroad. There is no FBR requirement for male migrant workers. We use the age of 5 as a policy cut-off to define treatment and control statuses. However, it should be noted that although females with children above 5 years old are able to migrate, they are also required to arrange a substitute caregiver to protect children.[4] This requirement may lead to an underestimation of the policy effects, as compared to a scenario where a clear comparison could be made between those with and without policy exposure. This is because our control households may also benefit from the policy, particularly through its effects on children, thereby diluting the measured impact of the intervention.

As intended, the policy, which was introduced in 2013, led to a sharp decline in female international migration departures, as shown in Fig. 1. In contrast, male migration continued to increase until 2014.

The FBR policy has been the subject of policy and academic discourse. Abeyasekera and Jayasundere (2015) critically analyze the FBR policy from a feminist perspectives, and the gender-specific nature of the policy has made it controversial (UN, 2015). By targeting female workers, the policy constrains their economic opportunities and may infringe on their rights.[5] Some existing literature has also examined the effects of the FBR policy quantitatively. Weeraratne (2016) finds that the policy negatively affected female foreign employment based on official departure statistics. In a subsequent study, Weeraratne (2021) further shows that the decline was mainly concentrated among low-skilled groups, which include domestic workers. By exploiting exogenous variation in

[4] The Circular also establishes minimum age requirements for migrants themselves, which vary by destination regions.

[5] The policy is reported to make female migrants vulnerable at the destination and induce some corruption in the process (Weeraratne, 2016, 2022).

the FBR policy, similar to the approach taken in this study, Peru (2023) finds that the policy's impact on fertility decisions varies by age and wealth of females. However, little is known about its causal effects on children. While not only Sri Lanka but also several countries in South and Southeast Asia have implemented or previously adopted similar restrictive migration policies with comparable objectives (Lenard, 2022), there has been limited causal analysis of their impacts. By providing the causal evidence on how such migration restrictions affect children left behind, this paper speaks to ongoing policy debates on the restriction of international migration, particularly in developing countries.

Child outcomes examined in this study include health and education. A relevant institutional background is Sri Lanka's provision of free universal healthcare and education to its citizens. The public healthcare system, funded by the government, ensures free access to hospitals, clinics, medications, and preventive programs. While public hospitals sometimes experience overcrowding, they remain the primary healthcare providers for the majority. Those who can afford it also have access to private hospitals for quicker service and specialized treatments. Similarly, primary education is free and compulsory from the age of 5 through 5th grade, followed by an additional four years of free and compulsory secondary education, as well as free public higher education. In addition to free tuition, the government supplies free textbooks and uniforms, ensuring widespread access to learning. As with healthcare, private schools also exist and charge tuition for those seeking alternative options. Together, these free public services play a crucial role in the country's social and economic development.

4. Data

We use repeated cross-sectional data from the Household Income and Expenditure Survey (HIES), conducted by the Department of Census and Statistics in Sri Lanka. The survey collects household-level expenditure data and individual-level income information, along with some demographic characteristics.

To evaluate the FBR policy, we use three survey waves: HIES 2009/10 and HIES 2012/13, which were conducted before the FBR policy,[6] and HIES 2016, conducted after the policy. Our main sample includes households with the youngest child aged 10 or younger. The age of 10 is the final year of primary education. We also restrict the analysis to households with the youngest child aged 2 or older, as the policy may influence fertility decisions as discussed by Peru (2023), which we will elaborate on further later. We also use a sample of individual children within this age range (i.e., 2 to 10 years old) to evaluate the policy impact on child outcomes.

Although the data do not provide information about migrants themselves (e.g., age and sex), we can identify whether a household sends a migrant and whether they migrate domestically or internationally. Additionally, for a subsample of households with a clear parent–child link (hereafter, the parent–child subsample), we can infer whether the mother is present at home, i.e., living together with the child.[7] Note that while this subsample is informative about mothers' responses to the policy, it may be subject to self-selection—and that selection could itself be influenced by the policy. To assess overall impacts, we therefore rely primarily on the full sample and present subsample results as corroborating evidence.

Table 1
Summary statistics of migration and household characteristics.

	Mean	[SD]
All sample: $N = 22419$		
Migration outcomes		
Any migrant abroad	0.08	[0.27]
Any remittance abroad	0.09	[0.29]
Amount of remittance abroad	14,200.79	[61882.21]
Any remittance domestic	0.08	[0.28]
Amount of remittance domestic	9340.43	[44393.19]
Family composition		
# of hh members incl. migrants	4.84	[1.41]
# of children 0–4 years old	0.47	[0.56]
# of children 5–9 years old	0.81	[0.64]
# of children 10–14 years old	0.56	[0.69]
Parent-child sumsample: $N = 17213$		
Mother present	0.97	[0.16]

This table summarizes household characteristics, including migration outcomes and family composition. The sample size is 22,419, except for "Mother present," which is available only for the "parent–child" subsample of 17,213 observations. Remittances are defined for the past 12 months.

Table 1 presents summary statistics for the migration and family composition variables. The sample pooling households with at least one child aged 2–10 from the 2009/10, 2012/13, and 2016 waves contains 22,419 observations. Note that the presence of the mother is only known for the parent–child subsample ($N = 17,213$, which corresponds to 77 % of the all sample households).

Eight percent of households in our sample have a migrant abroad. Nine percent of the sample households reported receiving remittances from abroad, with an annual average amount of 14,200 LKR. Remittances are relatively common: 9 % of households reported receiving remittances from abroad within the last 12 months, while 8 % received domestic remittances within the same period. The annual average amount is 9340 LKR, which is about two-thirds of the amount received from abroad.[8] The average household consists of 4.84 members, including migrants. On average, households have 0.47 children aged 0–4 and 0.81 children aged 5–9. In the parent–child subsample, 97 % of them have mothers at home.[9]

Table 2 presents the summary statistics of child-level outcomes of human capital investment. In our analysis, child health is measured by healthcare utilization by children, such as outpatient visits within the last month and inpatient stays within the past year. We also categorize outpatient visits and inpatient stays based on their underlying reasons such as check-ups and treatment for illness. We also analyze the presence of any chronic diseases. There is notable variation across different outcomes. On average, 33.6 % of children experienced any outpatient visit, primarily due to illness, while reported check-ups are relatively rare, with an average of only 0.8 %. Additionally, 6.4 % of children experienced inpatient care, with the majority of cases being illness-related. The

[6] HIES 2012/13 was conducted from July 2012 to June 2013. The FBR policy was announced in June 2013 and took effect in July 2013.

[7] The survey records the relationship between the household head and each member, except for migrating members. When a young household member (aged 10 or below in our analysis) is recorded as a child of the head, the mother is either the head or the spouse of the head, allowing us to infer whether she is at home. However, if the young household member is a grandchild of the head or a nephew/niece, their relationship is recorded as "other relative," making it impossible to identify their parent and, consequently, whether the mother is present at home.

[8] Remittance amounts are collected via the questionnaire item "current remittances and transfers," referring to the past 12 calendar months. Respondents report amounts separately for "outside the country" and "within the country," which we define as international and domestic remittances, respectively. The questionnaire does not identify the sender, so remittances cannot be linked to a specific household migrant; the sender may be someone other than the household's migrant.

[9] Appendix Fig. A.1 shows the relationship between the age of the youngest child and two outcomes: migration and the mother's presence, before and after the FBR policy. The share of households with any migrant abroad was lower for children aged 2–4 before the policy but increased afterward, while the probability of the mother being at home rose for households with children under 4 after the policy. At the age cutoff of 5, both effects diminish, indicating a neutralizing impact of the policy for older children.

Table 2
Summary statistics of child development outcomes.

	Mean	[SD]	N
Health			
Any outpatient	0.336	[0.472]	32,621
Outpatient for illness	0.323	[0.467]	32,621
Outpatient for check-up	0.008	[0.091]	32,621
Any inpatient	0.064	[0.245]	32,621
Inpatient for illness	0.050	[0.218]	32,621
Any chronic disease	0.035	[0.184]	32,621
Education			
School attendance	0.986	[0.117]	20,221
Grade retention	0.003	[0.054]	18,479
Current Grade	3.428	[1.576]	19,892
Age appropriate grade	0.975	[0.156]	19,892

Notes: This table presents the summary statistics of child outcomes. The sample is restricted based on age criteria: for health outcomes, the analysis includes children aged 2 to 10 years. For education-related outcomes, the sample is further restricted to children aged 5 to 10 years, as they are expected to be in school. Educational outcomes are well-defined only within this age range, with school retention specifically considered for children aged 6 to 10 years. The last three questions are only asked conditionally on her being attending school.

prevalence of chronic diseases is 3.5 %, which aligns with expectations given the population of young children.[10]

Apart from chronic disease, we use healthcare utilization to measure child health, whereas previous studies (e.g., Gosselin-Pali, 2025; Meng and Yamauchi, 2017) have commonly relied on anthropometric measurements such as height-for-age z-scores, which are not collected by HIES. While healthcare utilization primarily captures short-term and acute health conditions, anthropometric measures tend to reflect long-term nutritional status. Our study complements previous findings by examining child health from a different perspective. However, healthcare utilization requires a more nuanced interpretation, as it depends not only on a child's underlying health status but also on access to healthcare services. We will further discuss this when presenting and interpreting our results in Section 6.

While health outcomes are available for all the children in our analysis, education outcomes are only defined for children above 5 years as primary education starts at the age of 5 in Sri Lanka. We examine the following educational outcomes: school attendance, grade retention, current grade, and age appropriate grade.[11] Table 2 shows that primary education, which is both mandatory and free in Sri Lanka, appears to be highly effective—99 % of children attend school, the rates of grade retention are minimal, and 97 % of children are in an age-appropriate grade without any cumulative grade repetition. Consequently, our empirical analysis focuses on the relatively small margins of these outcomes.

5. Econometric strategy

We now turn to the empirical setup. Our central question is whether restricting mothers' migration ultimately affects human capital investment in children's health and education. We test this using a difference-in-differences design that compares households whose youngest child is below age 5 (treated) to those whose youngest child is age 5 or older (control), before and after the introduction of the FBR policy. We also examine potential mechanisms to interpret the overall effects, guided by Section 2.

Policy exposure is defined by the age of the youngest child at the survey date, since the FBR restricts migration for households with a youngest child under 5.[12] This definition captures current policy restrictions rather than the duration of exposure. Our DID specification uses this cross-sectional treatment contrast and the pre- vs. post-policy timing as the temporal dimension.[13]

We use the household as the unit of observation for analyzing migration and maternal presence, as well as for examining remittances, income, and labor substitution. The regression specification for the difference-in-differences analysis is as follows:

$$y_{ht} = \gamma_h + \lambda_t + \alpha(\text{Treated}_h \times \text{After}_t) + \mathbf{X}'_{ht}\beta + \varepsilon_{ht} \quad (1)$$

for household h at the time of survey $t \in \{2009/10, 2012/13, 2016\}$. Treated_h is a dummy variable equal to 1 for households with the youngest child aged below 5, and 0 otherwise. After_t is an indicator variable equal to 1 for the period after the introduction of the FBR policy ($t = 2016$). λ_t captures survey wave fixed effects and γ_h captures fixed effects for age of youngest child in household. We control for household characteristics $\mathbf{X}_{ht}$ (a school dummy, and family composition including the numbers of children aged 0–4 years, 5–9 years, and 10–14 years, and ethnicity, religion, and education of household head) and district fixed effects, sector (urban, rural, or estate) fixed effects, and survey month fixed effects, and ε_{ht} is the error term. We cluster standard errors at the district sector level. The coefficient of interest is α.

We also conduct the child-level analysis to estimate the effects of the policy on human capital investment. The regression specification is almost the same as Eq. (1), but the sample consists of children whose ages range from 2 to 10. The treatment variable Treated_h is still defined at the household level, meaning that a child is treated if they belong to a household where the youngest child is below the age of 5. This definition is motivated by our proposed main mechanisms, which suggest that the mother's presence and income effects are crucial for child outcomes and operate at the household level. In other words, policy exposure may benefit older siblings if they have younger siblings below the age of 5. That is, a child aged above 5 will have a value of 1 for this variable if they have a younger sibling under the age of 5. We include child characteristics such as sex, ethnicity and own age fixed effects in addition to the household characteristics. At the child analysis, we cluster standard errors at the household level.

The empirical approach leverages a natural experiment comparing households with the youngest child in different cohorts. Our identification relies on the parallel trends assumption—that households with the youngest child under 5 and those with the youngest child over 5 would have followed similar trends in the absence of the policy. With only two waves before and one after the policy, we cannot formally test

[10] In Appendix Fig. A.2, we present child-level health outcomes across different ages. Some of these outcomes are age-sensitive: we observe a clear pattern of monotonic decline with age for outpatient visits (both general and illness-related) and inpatient visits. Check-ups are more frequent at younger ages, while the prevalence of chronic diseases appears constant across all ages at very low rates.

[11] School attendance is a binary indicator of whether a child is currently attending school. Grade retention is also a binary indicator, defined as 1 if a child's grade in the current year is the same as in the previous year, and 0 otherwise. Current grade refers to the grade the child is currently enrolled in. Age-appropriate grade is a binary indicator of whether a child is enrolled in the grade typically expected for their age (e.g., Grade 1 for age 5, Grade 2 for age 6, and so on). Note that the last three outcomes are only defined for children who are attending school, and grade retention is well-defined by construction only for those above age 6.

[12] For instance, if a household has two children, aged 3 and 8, the mother is restricted from migrating under the policy because the youngest child is under 5 years old. This also effectively allows us to estimate the spillover effects on non-policy-targeted children—those who are above age 5—by comparing households with and without younger siblings.

[13] Note that as discussed in Section 3, households in our comparison group are partially affected by the policy though less restrictive. Thus, the estimated effects below are likely to underestimate the impact of restricting international migration compared to pure control households.

for pre-trends. However, we offer a contextual discussion supporting the plausibility of this assumption.

First, Appendix Table A.1 presents summary statistics for outcomes and family composition variables by treatment status (i.e., whether the household's youngest child is under 5), focusing on the pre-policy period (2009 and 2012 waves). We find no significant pairwise differences by treatment status. Household composition differs by design—control households have no children aged 0–4 but tend to have more older children.

Second, the policy may affect fertility decisions (Peru, 2023). With the new policy, having a child decreases economic opportunities abroad for a certain number of years. If fertility decisions are influenced by unobservable characteristics, the parallel trends assumption may not hold. For example, if those who are more passionate about child-rearing are less likely to be affected by the policy, the estimated DID coefficients will overestimate the true impact of the policy. To address this concern, the analysis focuses on cohorts that should not be affected by the policy in terms of fertility decisions. Specifically, children born after June 2014 (i.e., under the age of 2 in the 2016 survey) are likely to have been affected by the policy, whereas the decision to have a child born before that date (i.e., over the age of 2 in the 2016 survey) had already been made, and households could not have altered it in response to the policy. Therefore, we restrict the sample to households whose youngest child is over 2 years old.

Third, the policy's age cutoff closely aligns with the timing of primary school entry. Outcome trends may differ between preschool and school-aged children if school attendance influences migration decisions by reducing childcare burdens at home, though the direction of this effect is unclear. Therefore, our regression analysis explicitly controls for an education cohort dummy.[14]

In addition, there is a concern regarding the timing and exposure to the policy. For instance, a mother with a 5-, 6-, and 7-year-old child in 2016 is not currently restricted from migrating under the policy but were restricted when the policy was in effect three years earlier. This could alter their migration decision due to the earlier policy enforcement. Additionally, the policy may impact child health gradually rather than immediately. Children aged 5 to 7 in 2016 may have been influenced by the policy implemented in 2013. Including these children and their households in the control group may bias the estimated effects of the policy. We refer to this issue as "previously treated". We will test the impact of this issue on our findings as a robustness check later.

6. Estimation results

Following the conceptual framework in Section 2, we first show that the policy increases mother's presence at home and is associated with improvements in child health. We then examine an alternative channel—the policy's effects on remittances, household income, and intra-household labor reallocation. Finally, we assess impacts on children's educational outcomes.

6.1. Main results: effects on migration, mother's presence, and child health

Table 3 shows the DID estimates of mother's international migration restriction on mother's presence. The dependent variable of Column 1 is any household member migrating abroad while Columns 2 and 3 represent whether the mother migrates abroad and whether the household has the mother present at home. While Column 1 uses all the entire sample of households, Columns 2 and 3 use the parent–child subsample.

The results show an economically and statistically significant impact of restricting mothers' international migration on both the migration decision and mothers' presence at home. The estimated effect is a 1.5 percentage point decrease in the likelihood of any household member migrating abroad, statistically significant at the 5 percent level, compared to the control group of 7.6 %, which represents a 19.7 % decrease. Column 2 shows that the FBR policy significantly decreases mothers' international migration by 0.7 percentage points, which is substantial compared to the control group of 1.5 %. Column 3 reports a significant increase in mothers' presence at home. The result indicates that exposure to the policy increases mothers' presence by 1.2 percentage points from the control group of 97.4 %.[15,16]

Table 3
Impact of mother's international migration restriction on mother's presence.

	Any household member migrating abroad	Mother migrating abroad	Mother present at home
	(1)	(2)	(3)
Treated × After	−0.015**	−0.007*	0.011**
	(0.006)	(0.004)	(0.005)
Control mean	0.076	0.015	0.974
Sample	All	Parent-child	Parent-child
Observations	22,419	17,213	17,213

Note: The table presents DID estimates of the impact of restricting mothers' international migration on mothers' presence outcomes. The dependent variable in column 1 is whether any household member migrates abroad; column 2 is whether the mother migrates abroad, constructed by two conditions: whether the mother is not present and any household member migrates abroad; and column 3 is whether the mother is present at home. "Treated" is a dummy variable indicating that the household's youngest child is below age 5, and "After" is a dummy indicating that the survey wave occurred in 2016. All columns include fixed effects for the age of the youngest child and survey wave. Other control variables include a school dummy, and family composition (including the number of children aged 0–4 years, 5–9 years, and 10–14 years), ethnicity of the household head, religion of the household head, education of the household head, district fixed effects, sector fixed effects, and survey month fixed effects. Standard errors are clustered at the district-sector level. The row labeled "Control mean" indicates the average outcome for households whose youngest children were over age 5 before the 2013 survey. The row labeled "Sample" indicates the sample of households, where column 1 is restricted to households with the youngest children aged 2 to 10 years, while columns 2 and 3 further restrict the sample to 'Parent–child subsample' for whom a detailed parent– child relationship can be identified (See Section 4 for the definition). * denotes significance at 0.10; ** at 0.05; and *** at 0.01.

Appendix Fig. A.3 shows the event-study plots on any migrant abroad and mothers' presence displaying wave-specific treatment coefficients. The DID estimates shown above appear to be driven by the change between 2012 and 2016, which coincides with the timing of the introduction of the FBR policy in 2013.6, rather than capturing general trends or unusual events before the policy. We cannot reject the null hypothesis that the treatment coefficient in 2009 is equal to zero; indeed, the point estimates are very close to zero. After the introduction of the policy, there are significant point estimates in 2016, where we observe a significant drop in any migrant abroad and a significant jump in mothers' presence, as expected.

Table 4 presents the effects of restricting mothers' international migration on child health outcomes. We do not observe any significant effects on outpatient visits, although the sign of the point estimates aligns with the expectation of improvement in child health. There are negative but insignificant effects on outpatient visits for treatment for illness, and positive but insignificant effects on checkups. However, there

[14] Policy exposure remains in effect until a child turns 5, while primary education begins in January of the year after the child turns 5. This creates a gap that varies depending on the timing of the survey and the child's birth date.

[15] As outlined in the policy, control group households are also required to arrange a caregiver, which likely attenuates our results. Consequently, our estimates provide a lower bound of the true effect of restricting mothers' international migration.

[16] As shown in Column (1) of Appendix Table A.2, we also confirm that mothers' domestic migration does not increase.

Table 4
Impact of mother's international migration restriction on child health.

	Outpatient			Inpatient		Chronic disease
	Any	Illness	Check-up	Any	Illness	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × After	−0.009	−0.012	0.002	−0.011**	−0.008*	−0.000
	(0.012)	(0.012)	(0.002)	(0.006)	(0.005)	(0.005)
Control mean	0.345	0.333	0.007	0.072	0.057	0.035
Observations	32,621	32,621	32,621	32,621	32,621	32,621

Note: The table presents DID estimates of the impact of restricting mothers' international migration on child health. The dependent variables are dummy variables indicating outpatient visits for any reason, illness, and check-ups; inpatient visits for any reason and illness; and the presence of any chronic disease. "Treated" is a dummy variable indicating that the household's youngest child is below age 5, while "After" is a dummy variable indicating that the survey wave was conducted in 2016. Standard errors are clustered at the household level. All columns include fixed effects for the age of youngest child and survey wave. Other control variables include a school dummy, and family composition (including the number of children aged 0–4 years, 5–9 years, and 10–14 years), ethnicity of the household head, religion of the household head, education of the household head, own age fixed effects, district fixed effects, sector fixed effects, and survey month fixed effects. The row labeled "Control mean" indicates the average outcome for households where the youngest child was above age 5 before the 2013 survey. The sample is restricted to children aged 2 to 10 years, with at least one sibling aged 2 to 10 years. Clustered standard errors are at the household level. * denotes significance at the 0.10 level; ** at the 0.05 level; and *** at the 0.01 level.

is a significant decrease in inpatient stays for any reason, particularly for inpatient stays for treatment for illness, and these decreases are economically significant compared to the control mean. Any inpatient stay decreases by 1.1 percentage points relative to the control group of 7.2 %, representing a 15.2 % decrease. Inpatient stays for treatment for illness decreased by 0.8 percentage points compared to the control group of 5.7 %, representing a 14 % decrease. Finally, the introduction of the FBR policy does not appear to affect chronic diseases, likely because these conditions are too early to be diagnosed in the young children included in our analysis.[17]

These results appear to be driven by the increased presence of mothers at home. The policy leads to mothers remaining at home to care for their children, which, in turn, contributes to improved child health outcomes. This suggests that the policy as intended improves the child human capital development. The results are in line with the results by Meng and Yamauchi (2017), which demonstrate the adverse effects of parental, particularly maternal, migration on child nutritional outcomes.

However, we should interpret healthcare utilization carefully, as it is related to not only health conditions but also access to healthcare services. In contrast to our preferred interpretation, there is an alternative interpretation of the results. As discussed in Section 2, the restrictive migration policy leads to income decreases by reducing earning opportunities abroad and remittances from abroad. Due to these income reductions, healthcare services may become unaffordable. If this is the case, significant decreases in inpatient stays would not indicate improvements in child health; instead, they may merely suggest less access to healthcare services, without implying any actual change in children's underlying health conditions. However, we argue this interpretation is not plausible in the context of this study. First, as shown later, while the policy reduces remittances from abroad, this decrease is offset by an equivalent increase in domestic remittances, resulting in no significant change in household incomes. Additionally, Sri Lanka's free universal healthcare system, as explained in Section 3, minimizes the relevance of financial constraints in accessing healthcare. Therefore, our results suggest that the income channel is neutralized, and the observed decrease in impatient stays reflects an improvement in child health, which can be attributed to the increased presence of mothers at home.

Before discussing other mechanisms including such income channels, we check the robustness of our main results. Here, we provide four pieces of evidence to support the main findings.

First, we provide evidence that the issue of previously treated households, discussed in Section 5, may not affect our results in a substantial way. Appendix Fig. A.5 shows the youngest child's age-specific treatment coefficients for any migrant abroad and mothers' presence. The results for any migrant abroad seem to be driven by a decrease in the outcome for households with the youngest child aged 2–4, where the magnitude decreases as age increases, compared to the reference age of 5. However, we observe an increase or zero coefficients for children aged 6 or older. A similar (opposite sign) pattern is observed for mothers' presence. However, we also note that the effects for the age of the youngest child at 6 and 7 seem to move in the opposite direction, which may suggest some influence of the previous treatment.

Appendix Table A.3 also presents the results of an additional robustness check addressing the issue of the previous treatment. We exclude children from households with the youngest children aged 5–7, as they were exposed to the policy at its introduction but are currently classified in the control group. The results are consistent with those in Table 4: both the reductions in any inpatient stays and inpatient stays for treatment for illness are statistically significant at the 5 % level, with estimated decreases of 1.5 percentage points.

Next, we examine the effects on the parent–child subsample to assess sensitivity and the potential impact of sample selection bias.[18] Appendix Table A.4 presents the effects of restricting the samples to the parent–child subsample on child health outcomes. We observe similar coefficients for any inpatient stays and inpatient stays for treatment for illness, although the latter becomes slightly less precise.

Finally, we conduct falsification tests by redefining policy exposure to a different timing: treatment is defined for at the cutoff of ages 6 to 10, instead of the actual treatment age of 5. Appendix Fig. A.6 illustrates the placebo effects. Although we observe some significant effects at age 6 (and at age 7 for mothers' presence), likely due to the previously treated issue discussed above, we confirm that the coefficients are not statistically significant at the pseudo cutoff ages of 8 to 10.

[17] In Appendix Fig. A.4, we estimate the heterogeneous treatment effects by age and find that the effect on outpatient visits is significant and positive at age 4, while the effect on inpatient care appears to be driven by younger ages.

[18] The DID coefficient for being in the parent–child sample is negative and insignificant.

Table 5
Impact of mother's international migration restriction on remittance and income.

	Amount of remittance abroad (i.h.s)	Amount of remittance domestic (i.h.s)	Amount of total remittance (i.h.s)	Any remittance abroad	Any remittance domestic	Total household income (i.h.s)
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × After	−0.189*	0.196*	0.008	−0.014*	0.016	0.030
	(0.097)	(0.113)	(0.126)	(0.008)	(0.010)	(0.031)
Control mean	1.038	0.822	1.818	0.088	0.073	13.364
Observations	22,419	22,419	22,419	22,419	22,419	22,419

Note: The table presents DID estimates of the impact of restricting mothers' international migration on household remittance and income. The dependent variables are remittances and total household income. Columns (1), (2), (3), and (6) show the amounts of remittances from abroad, domestic, and total remittances, and total household income, respectively, all transformed using the inverse hyperbolic sine. Columns (4) and (5) indicate whether there are any remittances from abroad and domestic remittances, respectively. "Treated" is a dummy indicating that the household's youngest child is below age 5, and "After" is a dummy indicating that the survey wave was in 2016. All columns include fixed effects for the age of youngest child and survey wave. Other control variables include a school dummy, and family composition (including the number of children aged 0–4 years, 5–9 years, and 10–14 years), ethnicity of the household head, religion of the household head, education of the household head, district fixed effects, sector fixed effects, and survey month fixed effects. Standard errors are clustered at the district-sector level. The row labeled "Control mean" indicates the average outcome for households whose youngest children were over age 5 before the 2013 survey. * denotes significance at 0.10; ** at 0.05; and *** at 0.01.

6.2. Other possible mechanisms: income effects and intra-household labor substitution

Our main results above show that policy exposure increases mothers' presence by discouraging international migration, which appears to enhance child health. However, there is the possibility of other potential mechanisms that the policy could affect children as discussed in Section 2. Below, we examine whether the policy had a negative impact on income and how households adjusted their intra-household labor supply in response.

Table 5 presents the effects of restricting mothers' international migration on remittances and income (inverse hyperbolic sine transformed). We find that the inverse hyperbolic transformed amount of remittances from abroad decreases by 0.19 due to policy exposure at the 10 % significance level. On the other hand, policy exposure increases the inverse hyperbolic transformed amount of domestic remittances by 0.20. This likely reflects household coping behavior: since the FBR policy restricts only mothers' international migration and thereby caused a significant drop in remittances from abroad, households may have sought domestic remittances to compensate the loss in international remittance. Column (3) shows the effect on total remittances, and we do not find statistically significant effect, indicating the decreased remittance abroad appears to be offset by the increased domestic remittances.

Consistent with the policy's intent and the resulting decline in maternal migration, we observe a significant decline in the likelihood of receiving any remittances from abroad at the extensive margin. The FBR policy reduces this probability by 1.4 percentage points, as shown in Column (4). However, Column (5) shows no significant increase in the likelihood of receiving domestic remittances, despite its positive sign and a magnitude comparable in absolute terms to that in Column (4). This suggests that the observed increase in remittance amounts reported in Column (2) is driven by the intensive margin—higher remittances among existing recipients—rather than a broader expansion at the extensive margin. One possible interpretation is that domestic remittances increase in response to shocks (in this case, the restrictive migration policy), facilitated by pre-existing migration networks, as suggested by Batista and Vicente (2023) and Bettin et al. (2024).

Column (6) shows the effects on total household income. Interestingly, despite policy exposure decreasing the amount of remittances from abroad significantly, there are no significant impacts on total household income. The decrease in remittances from abroad appears to be offset by household coping responses, mainly through an increase in domestic remittances.[19]

Table 6 presents the impact of restricting mothers' international migration on intra-household labor reallocation and household composition. We find no evidence that policy restrictions targeting mothers leads to a substitution toward fathers' international migration nor by other household members. Column (1) shows that the policy did not affect fathers' presence at home, as they are not directly targeted by the policy. Column (2) shows no significant effect on the likelihood of fathers migrating abroad.[20] Column (3) shows no significant effect on the likelihood of domestic migration by any household member although the estimate is positive and its magnitude is comparable in absolute terms to the decrease in international migration shown in Column (1) of Table 3. Appendix Table A.2 further examines effects on domestic migration. While there is no impact on mothers' migration—consistent with the earlier finding of increased maternal presence—we observe an increase in fathers' domestic migration in the parent–child subsample. Columns (3) and (4) examine changes in household composition by gender. The number of female adults increases slightly, though the estimate is imprecise, which is consistent with mothers remaining at home.[21,22]

However, Column (5) suggests a significant increase in maternal involvement within the household. The FBR policy results in a 2.6 percentage point rise in the likelihood of females reporting housework as their main activity. Although the data do not provide specific details on the nature of housework, it likely includes childcare. This supports the interpretation that improved child health outcomes are driven by increased maternal presence and enhanced interaction between mothers and their children.

In summary, although policy exposure led to a decrease in remittances from abroad, this reduction appears to have been offset by an increase in domestic remittances, resulting in a neutral overall income effect. Taken together with the findings from the previous section, the policy has a positive impact on human capital investment, as reflected by the reduction in healthcare utilization. This improvement in child health is primarily driven by the increased presence of mothers at home, consistent with the policy's intended goal. However, these positive outcomes seem to be made possible through household compensation for lost income via increased domestic remittances.

[19] See Appendix Table A.5 for the effects on the detailed disaggregated composition of income.

[20] We also find no significant effect on the probability of fathers being at home.

[21] This potential underestimation is likely due to the policy design—specifically, households with the youngest child aged above 5 still requiring to arrange a caregiver when migrating. As a result, the policy's indirect effects may extend to these households, thereby attenuating the measured impact.

[22] We also find no significant effects on the number of working female or male adults.

Table 6
Impact of mother's international migration restriction on labor reallocation.

	Father present at home	Father migrating abroad	Any household member migrating domestic	Any female housework
	(1)	(2)	(3)	(4)
Treated × After	−0.014	−0.006	0.015	0.026**
	(0.013)	(0.006)	(0.012)	(0.012)
Controls	✓	✓	✓	✓
Control mean	0.867	0.035	0.106	0.674
Sample	Parent-child	Parent-child	All	All
Observations	17,213	17,213	22,419	22,419

Note: The table presents DID estimates of the impact of restricting mothers' international migration on household labor reallocation. The dependent variables are household labor allocation outcomes including whether the father is present at home, whether the father is migrating abroad, any household member migrating domestically, and any female doing housework. "Treated" is a dummy indicating that the household's youngest child is below age 5, and "After" is a dummy indicating that the survey wave was in 2016. Standard errors are clustered at the district-sector level. Control variables include a school dummy, and family composition (including the number of children aged 0–4 years, 5–9 years, and 10–14 years), ethnicity of the household head, religion of the household head, and education of the household head. All columns include district fixed effects, sector fixed effects, and survey month fixed effects. The row labeled "Control mean" indicates the average outcome for those whose youngest children were aged above 5 before the 2013 survey. The row labeled "Sample" indicates: 'All' includes households whose youngest child is aged 2–10, while the 'Parent–child subsample' includes households for which detailed parent–child relationships can be identified (see Section 4 for the definition). * denotes significance at 0.10; ** at 0.05; and *** at 0.01.

Table 7
Sibling spillover effects on child health.

	Outpatient			Inpatient		Chronic disease
	Any	Illness	Check-up	Any	Illness	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Direct (children who are the youngest or are above age five)						
Treated × After	−0.020	−0.021	0.003	−0.017**	−0.013**	0.000
	(0.013)	(0.013)	(0.003)	(0.007)	(0.006)	(0.005)
Control mean	0.371	0.358	0.007	0.079	0.063	0.035
Observations	23,254	23,254	23,254	23,254	23,254	23,254
Panel B: Indirect (children who are not the youngest and are above age five)						
	Outpatient			Inpatient		Chronic disease
	Any	Illness	Check-up	Any	Illness	Any
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × After	0.017	0.009	0.003	0.002	0.002	−0.002
	(0.021)	(0.020)	(0.004)	(0.010)	(0.008)	(0.009)
Control mean	0.282	0.273	0.005	0.056	0.043	0.034
Observations	9367	9367	9367	9367	9367	9367

Note: The table presents sibling spillover effects of restricting mothers' international migration on child health. Panel A shows the effects on the subsample of children who is the youngest in the household or under 5, while Panel B shows the effects on the subsample of children whose age is not the youngest the household or is 5 years or older. The dependent variables are dummy variables indicating outpatient visits for any reason, illness, and check-ups; inpatient visits for any reason and illness; and the presence of any chronic disease. "Treated" is a dummy variable indicating that the household's youngest child is below age 5, while "After" is a dummy variable indicating that the survey wave was conducted in 2016. Standard errors are clustered at the household level. Control variables include a school dummy, and family composition (including the number of children aged 0–4 years, 5–9 years, and 10–14 years), ethnicity of the household head, religion of the household head, and education of the household head. All columns include age fixed effects, district fixed effects, sector fixed effects, and survey month fixed effects. The row labeled "Control mean" indicates the average outcome for households whose youngest children were aged above 5 before the 2013 survey. The sample is restricted to children aged 2 to 10 years, with at least one sibling aged 2 to 10 years. * denotes significance at the 0.10 level; ** at the 0.05 level; and *** at the 0.01 level.

6.3. Extension: policy target and sibling spillover effects

Next, we distinguish between the direct effects of the FBR policy on target children and potential spillover effects on their non-targeted siblings. Although the policy directly targets children under the age of 5, the increased presence of mothers at home may also benefit older siblings in the same household, potentially improving their outcomes as well. It is important to note that the estimates presented above capture both direct and indirect effects by design. To separate these effects, we split the sample into two groups: (1) children who are the youngest in the household or are above age 5, representing the direct effect, and (2) children who are not the youngest and are above age 5, representing the indirect or spillover effect.

Table 7 presents this subsample analysis.[23] The findings indicate that our main results are primarily driven by the direct effects on children targeted by the policy target. Panel A shows that the direct effects on inpatient stays are statistically significant at the 5 % level, while Panel B shows no statistically significant spillover effects. These results suggest that maternal presence is especially important during the targeted ages (i.e., under 5) for child health, consistent with evidence that maternal

[23] We also conduct a subsample analysis using an alternative definition of spillover effects, where direct effects are defined for children who are the youngest in the household, and indirect effects for those who are not. The results remain very similar.

Table 8
Impact of mother's international migration restriction on child education.

		Conditional on attending school		
	School attendance	Grade retention	Current grade	Age appropriate grade
	(1)	(2)	(3)	(4)
Treated × After	0.000	−0.003*	−0.025	0.000
	(0.004)	(0.001)	(0.019)	(0.005)
Control mean	0.984	0.005	3.404	0.971
Observations	20,221	18,479	19,892	19,892

Note: The table presents DID estimates of the impact of restricting mothers' international migration on child education. The dependent variables are current grade and dummy for retention. "Treated" is a dummy variable indicating that the household's youngest child is below age 5, while "After" is a dummy variable indicating that the survey wave was conducted in 2016. Standard errors are clustered at the household level. Control variables include a school dummy, and family composition (including the number of children aged 0–4 years, 5–9 years, and 10–14 years), ethnicity of the household head, religion of the household head, and education of the household head. All columns include age fixed effects, district fixed effects, sector fixed effects, and survey month fixed effects. The row labeled "Control mean" indicates the average outcome for households where the youngest child was above age 5 before the 2013 survey. The sample is restricted to children aged 5 to 10 years, as school age begins at 5. Educational outcomes are well-defined only within this age range, with grade retention specifically considered for children aged 6 to 10 by definition. In the survey, grade retention and current grade are asked only for children who are attending school, and thus age-appropriate grade is also defined only for this group. Clustered standard errors are at the household level. * denotes significance at the 0.10 level; ** at the 0.05 level; and *** at the 0.01 level.

migration during the preschool period delays cognitive development (Bai et al., 2022) and with findings from other settings (Currie and Almond, 2011).

We extend the analysis to examine the effects on children's educational outcomes. Since these outcomes are only measured for children above age 5, we focus on estimating spillover effects on non-policy-targeted children by comparing those over age 5 with and without younger siblings under age 5. While school attendance is observed for all school-aged children, grade retention and current grade are only available for those attending school. By definition, grade retention is well-defined only for children above age 6.

Table 8 presents the results. Column (1) indicates that the FBR policy does not improve school attendance among non-policy-targeted children who have younger siblings under the age of 5.[24] Similarly, we do not find evidence that policy exposure affects the current grade of enrollment or the probability of being in an age-appropriate grade. It is worth noting that the mean values of these outcomes are already very high, reflecting the strong compliance with Sri Lanka's mandatory schooling, which may limit the scope for large observable effects. However, we do find a statistically significant reduction in grade retention in the current year. Policy exposure reduces grade retention by 0.3 percentage points, compared to a control group mean of 0.5 %—a 60 % reduction. Given the very low control mean, this improvement in educational status is likely concentrated among children facing more challenging educational environments.

This result aligns with our research design, which focuses on the current mothers' presence. The null results for current grade progression are consistent with the fact that our treatment only addresses relatively immediate effects. However, the observed reduction in grade retention is likely driven by the current presence of mothers. The interpretation of these findings, however, requires caution. Our findings of spillover effect on grade retention but not on healthcare utilization may reflect different channels affecting these outcomes.[25] While we emphasize the importance of mothers' presence and income effects, these may influence only specific educational achievements, particularly in settings where nearly all children attend school.

7. Conclusion and policy implications

International migration is an important economic opportunity in developing countries, but it can separate mothers from their children, potentially harming child development. This paper studies a unique policy in Sri Lanka that restricts mothers from international migration in order to protect children.

Our results suggest that the introduction of the FBR policy is effective in improving human capital investment. The policy successfully discourages mothers from migrating internationally, increasing their presence at home. We show that the policy leads to a decrease in any inpatient stays of child, particularly for treatment for illness, indicating improvements in child health. Despite the reduction in remittances from abroad, the overall income effect is neutral, as households compensate through increased domestic remittances. We also find a suggestive evidence of positive spillover effects on non-policy-targeted children's education, as reflected in reduced grade retention.

We believe our findings have broader relevance beyond the Sri Lankan context. Restricting mothers' international migration increases their presence at home, with evidence indicating positive effects on human capital investment, particularly in children's health and education. Given that migration restrictions are a realistic policy tool considered by other developing countries, providing causal evidence on their impacts is valuable.

However, caution is needed in generalizing these findings. In our context, domestic remittances helped offset the loss of income from abroad. But in settings with limited labor market access or weak remittance systems, the impact on child development may be ambiguous—or even negative—if income losses outweigh the benefits of increased maternal presence. These results highlight the trade-offs between the economic opportunities provided by international migration and the benefits of a mother's presence for child development. A key policy implication at the household level is that ensuring sufficient domestic labor opportunities is crucial to compensating for the loss of international remittances at the household level.

It is also important to note that there are additional concerns surrounding the policy. First, the loss of international remittances at the

[24] Among the children not attending school in the sample, the stated reasons of non-attendance were: disability or illness (22.3 %), unwillingness to attend or poor academic progress (17.2 %), financial problems (6.3 %), and other reasons, each accounting for less than 1 %.

[25] Appendix Table A.6 repeats the spillover analysis for healthcare utilization outcomes, restricting the sample to match that used in the education analysis. We find results similar to Panel B of Table 7: there is no evidence of spillover effects on health outcomes. This also reconfirms that our main results of health are driven by the policy-targeted children.

household level is estimated to be substantial decrease. This poses a concern for governments in developing countries, as remittances are a critical source of foreign currency acquisition and may have significant macroeconomic implications. Second, there is reported unintended negative consequences. Weeraratne (2016) documented that although the FBR was successful in restricting female migration for domestic work, it also promoted migration outside Sri Lanka's legal framework, often through visitor visas, thereby increasing workers' vulnerability at their destination. This vulnerability was further exacerbated as women resorted to corrupt practices to circumvent the FBR requirement by forging documents. Often, the costs of falsifying FBR documents were covered by sub-agents or licensed recruitment agents, which led to exploitation and abuse of potential female migrants during the recruitment process. Additionally, the FBR has been associated with delays in the recruitment process, creating further barriers for women seeking legal migration opportunities (Weeraratne, 2022).

While protecting children is an important policy goal, it is equally essential to safeguard the rights of female workers. This study focuses on one side of this trade-off. Ongoing policy evaluation and discussion are crucial to fully understand and address these competing goals.

CRediT authorship contribution statement

Takuya Hasebe: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yuma Noritomo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bilesha Weeraratne:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT 5 in order to enhance the readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We thank Chris Barrett, Nancy Chau, Brian Dillon, Johannes Haushofer, Shuhei Kainuma, Daichi Shimamoto, Masaru Nagashima, and Kazushi Takahashi as well as conference and seminar participants at ADEA-QUT 2025, JEA 2025 Spring meeting, SEHO 2025 meeting, AEDC 2025, IHEA 2025 Congress, ESWC 2025, EEA Congress 2025, 15th Asian HEA joint conference, GRIPS, Tohoku University, Nagoya City University, and Kobe University for their constructive comments. This work is financially supported by Sophia University Special Grant for Academic Research and Incentive Allowance for Dissemination of Individual Research Results, Grant-in-Aid for Research at Sophia Institute of Human Security, and the Joint Usage / Research Center, Institute of Economic Research, Hitotsubashi University. (Grant ID: IERPK2511).

Appendix A. Appendix figures and tables

See Table A.1–A.6 and Fig. A.1–A.6

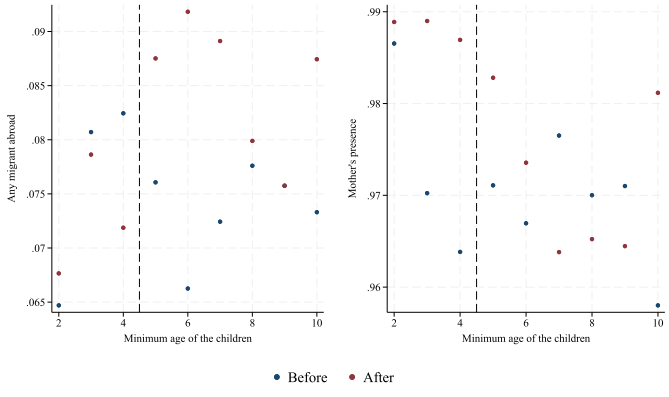


Fig. A.1. Migration and the age of the youngest child. Notes: these figures depict the relationship between the age of the youngest child in households and migration outcomes: any migrant abroad (left panel) and mother's presence (right panel). "After" refers to data from 2016, while "Before" refers to data from 2009/10 and 2012/13, indicating whether the data was collected before or after the introduction of the FBR policy.

Microcredit and willingness to pay for environmental quality: Evidence from a randomized-controlled trial of finance for sanitation in rural Cambodia

Ariel Ben Yishay · Andrew Fraker · Raymond Guiteras · Giordano Palloni · Neil Buddy Shah · Stuart Shirrell · Paul Wang

Journal of Environmental Economics and Management 86 (2017), 121–140 · doi:10.1016/j.jeem.2016.11.004

This reading edition reproduces the article's prose under CC BY 4.0 from the publisher full text archived in PubMed Central (PMC5669309). Tables, figures, equations rendered as images, footnotes, and references are omitted for compact printing; the remaining article text is not summarized or rewritten.

Abstract

Low willingness to pay (WTP) for environmental quality in developing countries is a key research question in environmental economics. One explanation is that missing credit markets may suppress WTP for environmental improvements that require large up-front investments. We test the impact of microloans on WTP for hygienic latrines via a randomized controlled trial in 30 villages in rural Cambodia. We find that microcredit dramatically raises WTP for improved latrines, with 60% of households in the Financing arm willing to purchase at an unsubsidized price, relative to 25% in the Non-financing arm. Effects on latrine installation are positive but muted by several factors, including a negative peer effect: randomly induced purchases by neighbors reduce a household's probability of installing its own latrine. On methodological grounds, this paper shows that a “decision-focused evaluation” can be integrated into academic analysis to provide insight into questions of general interest.

Introduction

Environmental conditions are typically worse in poor countries along key dimensions such as clean air and clean water (WHO, 2014b, WHO, 2014a, WHO, 2015), and the burden of environmental disease similarly falls disproportionally on poor countries (Prüss-Üstün and Corvalán, 2006). However, willingness to pay (WTP) for environmental quality is typically low in developing countries, and understanding the reasons why is a key research question at the intersection of environmental and development economics (Greenstone and Jack, 2015).

Poor sanitation in developing countries is an important example of this general phenomenon. Globally, 2.5 billion people live without access to improved sanitation, with one billion of these people practicing open defecation (WHO, 2014). Inadequate sanitation is believed to cause 280,000 deaths per year (Prüss-Ustün et al., 2014), contribute to serious health problems such as chronic diarrhea and tropical enteropathy (Dangour et al., 2013, Lin et al., 2013), and may diminish human capital through impacts on stunting and cognitive capacity (Spears, 2012, Spears, 2013). Although simple, relatively affordable solutions such as low-cost pour-flush latrines exist and major policy initiatives have promoted their adoption, growth in latrine coverage and reduction in open defecation have been slow in many parts of the developing world.

Several explanations for this puzzle have been proposed, including lack of information on health benefits, peer effects and social influence, supply-side failures, and difficulties in coordinating in the face of externalities and complementarities (Pattanayak, 2009, Pattanayak et al., 2009, Trémolet, 2011, Perez et al., 2012, Gertler et al., 2015, Guiteras et al., 2015b). In this paper, we focus on one aspect of a household's decision to purchase and install a latrine: latrines require a large up-front investment whose benefits are realized over time. This is a common characteristic of many actions required to improve environmental conditions and health, and could limit investment for any of several reasons (e.g., lack of consumer credit, high discount rates, or present bias) and suggests that interventions to fill missing capital markets may increase investment and improve welfare. However, there is little evidence on the effectiveness of credit-based interventions at affecting WTP for environmental quality (Greenstone and Jack, 2015), and recent research showing limited impacts of micro-credit interventions on income and welfare has led to increased skepticism of the value of micro-finance (Banerjee, 2013, Banerjee and Karlan, 2015).

This paper reports the results of what is, to our knowledge, the first randomized-controlled trial of the effect of micro-loans on WTP for improved sanitation. In a representative sample of 30 villages in rural Cambodia, our NGO partner conducted group information sessions and sales meetings to market a low-cost, hygienic latrine. In 15 randomly selected villages, households were offered the option of financing their purchase with a loan from a local micro-finance organization. In the remaining 15 villages, sales were made on a standard lump-sum, cash-on-delivery basis. To maximize the precision of our estimates and broaden the set of research questions we are able to address, we used the Becker-DeGroot-Marschak (BDM) mechanism to obtain precise measures of WTP for each household (Becker et al., 1964).

We find that the offer of a micro-loan dramatically increases WTP. Mean WTP in the Financing arm is $51.8, as compared to $29.9 in the Non-financing arm. This large increase in WTP occurs across quantiles and among both the poorest and relatively less-poor households.1 At the approximate break-even price of 40, only 25% of households agree to purchase the latrine without financing, but more than 60% of households offered financing are willing to pay this full, unsubsidized price. Because NGOs and social enterprises face large village-level fixed costs in marketing and delivery, these increases in WTP can increase the cost-effectiveness of interventions, even net of the cost of providing financing, by amortizing these fixed costs over a greater number of sales.

The impact of finance on latrine coverage, while positive, is not as large as the effect on WTP. Latrine installation rates 1.5–2 years after the sale were low in both arms, muting the effect of finance. The primary barrier to installation was the high cost of households' desired latrine

superstructure (walls and roof), for which financing was not offered. A secondary barrier, which we identify using the quasi-random variation in latrine purchases produced by the BDM mechanism, is a negative social spillover: exogenous increases in neighbors' purchases lead to lower installation rates and, ultimately, lower latrine coverage. This is consistent with several potential mechanisms: shared use of the latrines creating negative strategic complementarities in latrine installation – i.e. the private return to installation is decreasing in neighbors' latrine ownership - as well as with strategic substitutability in health investments.

This paper provides a model for the integration of “decision-focused evaluations” with academic analysis of general-interest questions. A decision-focused evaluation is one demanded by an implementer, designed to rapidly inform a specific policy or programmatic decision, and carried out in the specific context of interest and within in the implementer's usual operating and decision-making structures (Shah et al., 2015).2 The randomized evaluation in this paper was designed to inform our implementation partner's decision of whether it should scale up microfinance loans for latrines in rural Cambodia. The evaluation took 3.5 months to complete from inception to reporting results, cost less than 60,000 USD, and provided a clear, actionable recommendation to the implementer. Given the decision-focus of this evaluation, the survey instruments used were designed to minimize the time and cost of the evaluation, focusing only on collecting data that would be necessary for guiding iDE's decision. This focus limits our ability to investigate mechanisms behind the effects we observe, particularly our ability to distinguish credit constraints from present bias or high discount rates as the source of the effectiveness of the financing offer. At the same time, we are still able to provide rigorous evidence on an important question of broad interest.

The paper is organized as follows: in Section 2, we describe our setting and experimental design, including our study sample, intervention, and data collection. In Section 3, we show the dramatic effect of finance on demand. In Section 4, we show that the effect on coverage was less and explore reasons for this gap. In Section 5, we discuss the results, including the implications for cost-effectiveness, and conclude.

Experimental setting and design

We conducted our experiment in rural Kampong Thom province, Cambodia, a region with generally low access to improved sanitation: as of 2012, 31% of rural residents had access to a hygienic latrine. At baseline, 71% of respondents in our sample primarily defecate in the open, with 90% of children under 5 in these households doing so.3 These children experience diarrhea regularly, with 39% of children in sample households having diarrhea in the 7 days preceding the survey.

Formal financing is not uncommon, with 41% of households having held a loan from a bank or microfinance institutions in the preceding year. Consumer loans from formal sources are very rare; most loans are for productive assets only. Informal credit is more prevalent (61% of households), but average loan holdings are relatively small.

We partnered with two institutions to implement the sanitation marketing and the microfinance loans. iDE Cambodia (iDE) has conducted sanitation marketing and training of sanitation suppliers in rural Cambodia since 2007 and currently is active in sanitation in eight provinces across Cambodia. Microloans were provided by VisionFund Cambodia (VFC), a microfinance institution established in 1994 that has served more than 140,000 clients with a total loan portfolio over US $37M.4 iDE had worked in Kampong Thom province, but not in any of the sample villages, while VisionFund was active in all.

As a demand-driven, decision-focused evaluation, the study design was tailored to the implementers' budgetary, timeline, and operational constraints and the intervention mimicked the implementers' standard procedures as closely as possible. However, all data were collected by employees of IDinsight, independently of the implementers.

Sample frame and randomization

The study sample consists of 1,500 households from 30 villages in Kampong Thom province, selected using a multi-stage random sampling process. The initial village-level sample frame consisted of all 786 villages in Kampong Thom. Villages were then excluded from the sample frame on the basis of two criteria. First, to facilitate the selection of a representative sample, we dropped villages above the 95th percentile and below the 5th percentile with respect to population and share of households classified as “IDPoor.”5 Second, we excluded villages targeted by the Asian Development Bank's concurrent Rural Water Supply and Sanitation Sector Project, which provided heavily subsidized or free sanitation facilities to 75% or more of households. From the remaining villages, samples of 30 were randomly selected without replacement and with a probability of selection weighted by village size (number of households). At the end of each draw, we calculated the mean size, poverty rate and latrine coverage rate (measured at the district level) for the sample and compared these to the averages for Kampong Thom. If the sample differed significantly from the population (Kampong Thom as a whole), we discarded that sample. We repeated this procedure until it produced 100 qualified samples. Of these 100 qualified samples, one was randomly selected as binding. From this sample of 30 villages, 15 were randomly assigned to receive the Financing treatment.

In each village, a census was taken to obtain the names of the head of household and spouse and IDPoor status of the household, and to identify whether each household owned a latrine. Fifty households (average village size was 194 households) in each village were invited to participate in a group information session and sales meeting. These 50 households were randomly selected from the village population without a latrine, after stratifying on IDPoor status so that 30% of the selected households (i.e., 15 in each village) were classified as IDPoor. In four villages, fewer than 50 households did not have a latrine. In these four villages, all non-latrine-owning households were invited, yielding a total sample size of 1,383 households. On average, 76% of invited households attended the sales meeting. The household could be represented by any household member older than 18 with the authority to make large purchase decisions for that household. Field staff then followed up with non-attending households to conduct the same information and sales session, typically within one day of the initial session in the village. Ultimately, 1,380 of 1,383 invited households (99.8%) participated. Households were not compensated for attending the sales meeting, nor for any aspect of participation.

As described below, the intervention involved randomization of latrine prices at the household level. This randomization was unstratified and thus produced village-level variation in the average price draw.

Group information session

The core of the intervention was the group information session and sales meeting.6 Invited households gathered in a common location (e.g. school, village pagoda, village chief's house). Sales staff led a 45–60 minute interactive session emphasizing the health and convenience benefits of having a latrine and its status as an aspirational good. The latrine offered for sale included three concrete rings, each 80 cm in diameter, a concrete pan, a concrete slab with a porcelain bowl that fits into the pan, and a PVC pipe that connects the three rings to the concrete pan. The approximate cost of this set of parts was 160,000 KHR (USD 40).7 Participants were not given information on the cost. The latrine was marketed as being easy to self-install, requiring approximately one day's labor to dig a 1.5×1 m cylindrical pit to house the three concrete rings and a separate mound to keep the basin. No additional material was required to install the latrine after delivery other than a shovel and water to mix the mortar. The sale did not include installation or a superstructure.8

Financing treatment and sale

At the end of the information session, attendees were offered an opportunity to purchase a latrine. The sales offer was made using the Becker-DeGroot-Marschak (BDM) mechanism (Becker et al., 1964). In the Non-Financing (cash-on-delivery) arm, households were commonly anchored on an initial price of 300,000 KHR (USD 75), with prices subsequently reduced in 10,000 KHR (USD 2.5) increments until the participant was willing to accept that price. Prices were then increased in smaller increments (2,000 KHR) until the participant was no longer willing to pay. This mechanism identified the maximum WTP, or “bid”, for the latrine. The enumerator then allowed the household to choose from a set of sealed envelopes marked only with the participant's ID, each containing a randomly chosen price. The distribution of prices was 80,000 KHR, 120,000 KHR, 160,000 KHR, and 200,000 KHR (USD 20, 30, 40 and 50) with probability weights 1/4, 1/4, 1/4, 1/4 in the non-financing arm and 1/20, 9/20, 1/4, 1/4 in the financing arm, respectively.9,10 If the price inside the chosen envelope was less than or equal to the subject's bid, then the subject purchased the latrine at the randomly determined price. If the price in the envelope was greater than the subject's bid, then the subject could not purchase the latrine. The subject was not allowed to change her bid after the price was revealed. For expected utility maximizers, the subject's best strategy is to bid her maximum WTP.11 The bids were given and the price revealed in private. All subjects also took part in a practice round in which participants bid for a token item (a box of cookies worth approximately KHR 2,000) to ensure they understood the procedure.

The study of credit constraints involved two treatment arms: Lump Sum (control) and Financing (treatment), which were randomly assigned at the village level (as discussed above). In Lump Sum villages, the household was required to pay the full agreed price upon delivery, which would occur within 10 days. In Financing villages, the household was offered a loan from VFC, which could be repaid over a term of up to 12 months.12 The loans were group liability with monthly interest rates of approximately 2.8%.13 The sales mechanism was the same in Lump Sum and Financing villages, with two exceptions. First, in Financing villages, the loan option was explained before the bidding so that subjects could take this information into account when deciding their maximum WTP. Second, as indicated above, in Lump Sum villages, bids were made in terms of the full payment amount, while in Financing villages bids were made in terms of the monthly installment payments.14 Finally, in Financing villages, typically within 24 h of the group sales meeting, winning households met with a VFC underwriter, who used a basic battery of questions on the customer's age, income and assets to determine his eligibility for a loan. The underwriter then determined whether or not to extend the loan.15

To avoid non-random selection into attendance at the sales meeting, the treatment status of villages was not announced until the meeting itself. In fact, in Finance villages, no mention was made of the possibility of finance until the sales meeting, and in Lump Sum villages, the possibility of a loan was not mentioned at all. Households who were invited to the group meeting but did not attend were visited by enumerators within one day and offered an opportunity to participate at home. If they agreed, the information session and sales exercise were conducted in similar fashion to the group meeting.16

Data collection

We collected data using four instruments. First, we administered a census to all households in each village, obtaining the name of the head of household and spouse, whether the household owned a latrine, and whether the household was classified as IDPoor. Second, a baseline survey was conducted with all invited, consenting households. The survey covered latrine type and conditions, defecation practices, knowledge of latrine components and costs, household demographics, informal borrowing of rice or money from other people, housing quality, income sources, land holdings and agricultural production.

Third, we also obtained data from the sales exercise. For household h in village v, BDM provides data on willingness to pay (WTPhv), the price offer (Drawhv), and whether the household won the latrine (Wonhv=1{WTPhv≥Drawhv}). We also record whether the household actually purchased the latrine (Boughthv). If the household won the latrine but reneged, that is recorded as a refusal: Refusehv=(Wonhv=1)■(Boughthv=0). For households that purchase the latrine, the price paid is the price offer, Drawhv. The price paid is not defined for households that do not purchase. We also document whether households that lost attempted to bargain for the latrine and whether ex-post they wish they had bid more. For Financing villages, we convert to the declining balance sequence of payments to NPV using the interest rate of 2.8% per month as a proxy for VFC's cost of funds.

Finally, we conducted a follow-up survey to assess installation and use of the latrine via both self-reports and direct observation. The survey revisited all participating households approximately 18–24 months after the initial sales offers. Enumerators observed whether latrines (from any source) were installed, as well as conditions of installation, presence of a superstructure, and indicators of regular use and maintenance.

Effect of finance on demand

Main results

We are interested in the effect of finance on the (inverse) demand curve, i.e., s(WTPhv≥p), the share of households willing to purchase at a set of prices pL,…,pH. Fig. 1a plots s(WTPhv≥p) at each price p={5,10,…,100} separately by treatment group (Non-Financing and Financing). Households in both groups purchased latrines at relatively high rates (>80%) when the price is below 20 USD. Demand is quite elastic over prices in the [20,40] range (or 50–100% of market prices), especially in the Non-Financing group where only 27.9% of households would purchase the latrine at 40 USD.

In Fig. 1b, we plot the estimated treatment effect of finance, with 95% confidence intervals constructed using both randomization inference and standard regression methods. (We discuss randomization inference in Section 3.2 and in Aglasan et al., 2016) By randomization, this estimated treatment effect is the observed difference between the two demand curves in Fig. 1a. We weight each household equally but the effects are similar if we weight villages equally. We find large treatment effects that initially increase with price, reaching a maximum that exceeds 40 pp at 45 USD. Beyond this price, demand falls relatively quickly even in Financing villages, with treatment effects diminishing (although remaining greater than zero even at 100 USD). The lower bound for treatment effects exceeds 0 for all p≥20. At the approximate break-even or unsubsidized price of 40 USD, treatment effects are 33.8 pp (RI p-value=0.00), more than doubling the share of households purchasing latrines relative to the Non-Financing group.

One reason that financing may lead to higher purchase rates may be that households expect a substantial likelihood of defaulting. However, using administrative data from VisionFund, we track repayment rates for the ensuing year after the initial offer and find 100% repayment of the latrine loans.

Estimation and inference

By virtue of randomization, obtaining point estimates for treatment effects is simple: at each price p, we can take the difference in shares purchasing, i.e.,

Alternatively, we can use the estimated coefficient from a regression of 1{WTPhv≥p} on an indicator for whether household h's village was in the Financing treatment group, plus a constant.

Computing p-values and confidence intervals is not as straightforward. Because the treatment was randomized at the village level, and determinants of demand are likely correlated within village, inference must be made robust to clustering within village. However, because we have only 30 villages, the customary cluster-robust regression standard errors may be unreliable (Cameron et al., 2008).

Instead, we employ randomization inference both to obtain p-values and to construct confidence intervals by inverting the relationship between effect and p-value. While randomization inference for p-values has grown in popularity in recent years,17 accurate confidence intervals are important for policymakers because policy decisions usually depend on the magnitude of an effect, not just whether an effect exists, and use of randomization inference to construct confidence intervals is rare in economics.18 The intuition for randomization inference dates back to Fisher (1935): the researcher specifies a sharp null hypothesis, imposes this null hypothesis on the data, generates the distribution of the test statistic of interest under this null over all or many combinations of treatment assignments, and then compares the actual, observed value of the test statistic to its generated distribution to estimate how extreme the observed value is when the null is true (Rosenbaum, 2010, Imbens and Rubin, 2012). This procedure provides p-values directly when the null hypothesis is one of no effect. To obtain confidence intervals of level α, we test a series of null hypotheses (in our application, effects of –0.40,–0.39,…,+0.80) and construct the confidence interval as all values that are not rejected at the α level. We provide an extended discussion and formal exposition in a companion paper (Aglasan et al., 2016), and our code is available to interested researchers upon request.

For comparison, we also present “analytical” CIs computed using Huber-Eicker-White (HEW) heteroscedasticity-robust standard errors. We collapse the household-level data to the village-level share of households purchasing at price p, so, with one observation per village, clustering is not needed. However, this creates the possibility of small-sample bias in the HEW standard errors, so, following Angrist and Pischke (2009), we take the maximum of i.i.d, HEW, HC2,HC3, where HC2 and HC3 are finite-sample corrections to HEW standard errors. As shown in Fig. 1b, the randomization inference confidence intervals are only slightly wider than the “analytical” CIs.

Robustness checks

Effects on latrine installation

The previous section provides strong evidence that finance increases demand for sanitation. However, increasing initial purchases of latrines is necessary but not sufficient for improvements in environmental quality: toilets cannot reduce fecal loads in the environment unless they are installed and used. We conducted a followup survey 18–24 months after the initial sale to measure installation rates and collect objective indicators of latrine maintenance and use. At followup, we found that roughly 30–40% of households that had purchased a latrine during the sales exercise had installed it.21 Although this is comparable with rates of increased coverage from other sanitation interventions,22 it is somewhat surprising that only 30–40% of households that had purchased a latrine (as opposed to simply being encouraged to install one or given one for free) would have installed it nearly two years later. In this section, we first describe installation rates as a function of WTP and Financing treatment, then provide evidence of the cost of the superstructure as a barrier to installation, and finally explore the role of peer effects in depressing installation.23

Finance, WTP, and installation rates

Fig. 6a plots, at each indicated price, the installation rate among all households that purchased the latrine and had WTP greater than or equal to that price. That is, the figure answers the question, “If an NGO offered latrines for sale at a given price, what share of latrines purchased at that price would be installed?” The figure is restricted to prices where 10% or more of the households purchased a latrine ($50 or less for Non-financing; $85 or less for Financing). There are three salient messages from the figure. First, installation rates are below 40% except among the Non-Financing households with WTP above $40. Second, installation rates at any given level of WTP are slightly higher among Non-Financing households than Financing, although these differences are not statistically significant (see Fig. 6b). Third, there is little evidence of screening effects: installation rates are roughly constant as a function of WTP, although there are slight differences in the right tails of the WTP distribution: installation rates are slightly increasing among Non-Financing households with relatively high WTP and slightly decreasing among Financing households with relatively high WTP.24

Fig. 6 shows installation rates conditional on purchase, but it is also useful to consider unconditional installation rates. That is, if latrines were offered for sale at a given price, what share of households would purchase and install a latrine? Because of the randomness of BDM, this unconditional installation rate has to be calculated indirectly. We would like to calculate

However, the purchase decision, s(Purchase), has a random element since it depends in part on the BDM draw. To estimate the share that would purchase and install a latrine if the fixed price were p, we substitute s(Bid≥p) for s (Purchase). That is, we use the installation rate that would have occurred among participants if a fixed price of p had been offered (or if everyone bidding p or more had won).

This synthetic installation rate is plotted in Fig. 7a as a function of price, separately for Financing and Non-Financing villages. Fig. 7a answers the question, “If an NGO offered latrines for sale at a given price, what share of households would purchase and install a latrine?” Note that the installation rate is now higher for Financing villages than Non-Financing at most prices, seemingly contradicting Fig. 6. However, the two figures are in fact consistent – as shown in Fig. 6, installation rates are higher in Non-Financing villages holding WTP constant, but WTP is much higher in Financing villages (see Fig. 1), so overall coverage rates are generally higher. This difference is statistically significant at all prices above $40, as shown in Fig. 7b.25

What explains these installation rates? First, the cost of the superstructure (walls, roof, and door) appears to have been an important barrier. Although is possible to build an inexpensive superstructure using locally gathered materials (bamboo or thatch) or tin (typically less than $10), households exhibit a strong stated preference towards much more elaborate, expensive concrete structures costing $200 or more.26 In informal discussions during the followup, households that had purchased a latrine but not yet installed it commonly stated that they intended to install the latrine eventually but had not yet saved enough for their desired superstructure. This suggests two approaches that might be effective at increasing installation rates: first, encouraging households to construct acceptable interim superstructures while they gather funds for the high-end superstructure they desire; second, by financing the superstructure in addition to the latrine itself. The former approach could be implemented by a subsidy, e.g., forgiving part of the loan if the latrine is installed. Cash-on-hand constraints do appear to matter to some extent: using the random variation in price paid generated by the BDM draw, we find that each $10 reduction in the price paid (holding WTP constant) by the household increases the probability of installation at followup by approximately 4% points. See Appendix D for the details of this analysis.

Peer effects and installation

A second possible explanation for the low installation rate comes from a negative peer effect. Adoption of health technologies by nearby households has been shown to powerfully alter one's own beliefs about and propensity to adopt the technology (Godlonton and Thornton, 2012, Grant Miller and Mushfiq Mobarak, 2014). Moreover, peer effects may arise through varied mechanisms, some of which induce greater own adoption (learning, imitation) and others reduce it (sharing).27 The net effect of adoption by peers is thus ambiguous, and we seek to assess its sign and magnitude. Peer effects are typically difficult to identify, because observed correlation in behavior could be the result of a true, causal peer effect or simply homophily: peers tend to have similar tastes, so we might expect their behaviors to be similar even absent a true, causal effect (Manski, 1993). However, here we can use the randomization embodied in BDM to identify peer effects in installation. Because the price draw in BDM is random, there will be some random variation in latrine purchase rates in households’ peer groups. Furthermore, since BDM provides data on the household's WTP and the WTP of peer households, we can control for these and use only random variation in purchase rates conditional on own and peer WTP.

Ideally, one would collect detailed social network data to observe each household's relevant peer group precisely (BenYishay and Mushfiq Mobarak, 1964, Barrios et al., 2015). However, this was not feasible in this “decision-focused evaluation” because of time and budget constraints. Instead, we define each household's peer group as the other participating households in that household's village. Given the small size (on average, 250 households and 2–5 km diameter) and rural nature of the villages in our sample, it is plausible that the resulting village groups are not far from the true peer groups relevant for latrine ownership, installation, and use.

Our goal is to estimate the causal effect of latrine purchases by a household's peer group on that household's propensity to install a latrine. That is, we wish to estimate

where 1{Installh,v} is an indicator for whether household h in village v has installed a latrine at the time of the followup survey, and s(Buy~h,v) represents the share of other households (excluding household h) in village v who purchased a latrine. However, OLS estimation of Eq. (3) may produce biased estimates of the causal parameter of interest β1. The most plausible source of bias would be if households' preference for improved sanitation tended to be correlated within village, so in villages where WTP was greater and, therefore, s(Buy~h,v) tended to be higher, households were also more likely install a latrine they had purchased (i.e., εh,v was higher on average).

To overcome this identification problem, we can use the randomness provided by the price draw in BDM. The simplest strategy would be to instrument for s(Buy~h,v) with Draw¯~h,v, the average price draw among household h's peer group. We can increase power by using our knowledge of the distribution from which prices were drawn to define each household's chance of winning a latrine based on their bid. For

example, for a household with a sufficiently low WTP for the latrine (i.e., below the minimum possible price), the realized price draw will not have any predictive power for whether the household purchased the latrine. By focusing on the randomly induced difference between actual latrine purchases and those predicted by the households' WTP, we use only the price variation that can affect latrine purchase. Households with high WTP will be expected to purchase the latrine under most potential price draws, while the opposite is true for households with low WTP. The effect of price draws are thus conditional on WTP.

We can thus construct the "unexpected" purchases among each household's peer group, defined as $\tilde{s}(Buy_{\sim h,v}) = \sum_{j \neq h}[(Buy_{j,v}) - (\hat{Buy}_{j,v})]$, where $\hat{Buy}_{j,v}$ is household j's probability of purchasing given its WTP. We use the empirically observed distribution of price draws and the elicited WTP for each household to calculate their ex-ante probability of winning the latrine. As described in Section 2.3, latrine prices were drawn from a discrete distribution of $20, $30, $40, $50, with probabilities 1/4, 1/4, 1/4, 1/4 in the non-financing arm and 1/20, 9/20, 1/4, 1/4 in the financing arm, respectively.28 Therefore, any household with WTP<20 had a 0% chance to win the latrine, any household with 20≤WTP<30 had a 1/4 chance to win in the non-financing arm and a 1/20 chance to win in the financing arm, households with 30≤WTP<40 had a 1/2 chance to win, households with 40≤WTP<50 had a 3/4 chance to win, and those with 50≤WTP had a 100% chance to win. We can then calculate expected latrine purchase counts at the village level by summing these probabilities over all other participating households in the village.

Table 2 displays summary statistics for unexpected latrine purchases, $\tilde{s}(Buy_{\sim h,v})$, as well as the observed latrine installation rates at endline. On average, households are exactly as likely to have won a latrine through BDM as their WTP and the ex-ante distribution of potential draw prices would predict (mean and median of unexpected latrine ownership is 0.00), and the range of observed values is −0.75 to 0.95. In words, a value of unexpected latrine purchase of −.75 implies that a household in the non-financing arm bid between [40 and 50 USD), but received a price draw of 50 and therefore did not win the latrine. A value of unexpected latrine purchase of.95 implies that a household in the financing arm bid between [20 and 30 USD) and drew a price of 20 USD, winning the latrine in spite of their relatively low WTP for the latrine. Higher values of unexpected latrine purchase are found for households that were especially lucky to win the latrine through BDM given their stated WTP.

Although we can simply estimate the effects of peers' unexpected purchases on latrine installation, we want to maximize precision by controlling for a household's own predictors of installation, including own latrine purchase. We do so by controlling for the randomly induced latrine purchase based on a household's own price draw and WTP. That is, we construct the own-household equivalents of our peer measures as $\hat{Buy}_{h,v} = Pr(Draw \leq Bid_{h,v})$ and $\tilde{Buy}_{h,v} = Buy_{h,v} - \hat{Buy}_{h,v}$. The latter term, $\tilde{Buy}_{h,v}$, represents the unexpected or random component of latrine ownership.

Using these variables, we estimate

by OLS, with results reported in Table 3. The covariates $x_{h,v}$ are an indicator for whether the village was a part of the Financing treatment, the household's WTP in USD, and the village mean WTP in USD excluding the household's own WTP. Rather than base inference on asymptotic cluster robust methods, we estimate p-values and confidence intervals for each parameter of interest through cluster robust bootstrap procedures: the Wild bootstrap (Cameron et al., 2008) in OLS specifications and the Score bootstrap (Kline and Santos, 2012) in IV specifications. We use cluster-robust bootstrap methods instead of randomization inference primarily because the literature on randomization inference for IV (e.g. Rosenbaum, 2002, Small et al., 2008) focuses on a binary instrument and extending the methods to multi-valued instruments, as in the case of the price draw, was beyond the scope of this paper. Since bootstrapping and randomization inference produced very similar results in our other analyses (purchase shares, mean WTP, WTP quantiles), we are comfortable using only the bootstrap for inference in our analysis of peer effects.

As expected, unexpected own purchase has a positive and significant effect on the likelihood that a household installed a latrine; the point estimate suggests nearly a 15% point increase in the probability of latrine installation. The fact that the estimate is not 1 is indicative of the relatively low installation rates in our sample. More interestingly, the coefficient on the village leave out mean unexpected latrine share is negative and statistically significant ($p<.05$). This suggests that, after removing the effect of WTP and conditional on own ownership, having more neighbors purchase a latrine actually reduces the likelihood that a household installs a latrine. The point estimate indicates that shifting a household to a village with an unexpected latrine share that is one standard deviation (.045% points, or ≈2.5 additional households in a typical village with 50 participating households) higher is expected to reduce the likelihood that household installs a latrine by 4.1% points, as compared to the mean installation rate of 35% among purchasers.

This reduced-form estimate identifies the effects of unexpected peer purchase on the likelihood that a household has installed a latrine at endline. One may naturally want to assess the effects of any peer purchase (rather than only unexpected purchases); that is, to estimate effects of $s(Buy_{\sim h,v})$ rather than $\tilde{s}(Buy_{\sim h,v})$. We thus also estimate an instrumental variables (IV) specification in which we instrument for $s(Buy_{\sim h,v})$ with $\tilde{s}(Buy_{\sim h,v})$ and $Buy_{h,v}$ with $\tilde{Buy}_{h,v}$. The results, shown in Table 4, include point estimates that are precisely estimated and statistically indistinguishable from the OLS results. Once again, peer purchase dramatically reduces a household's installation of its own latrine. The first stage regressions show that unexpected peer purchase significantly predicts peer purchase and that unexpected own purchase significantly predicts own purchase.

Finally, Table 5 replicates the OLS specification from Table 3 but includes interactions between the Financing indicator and own and peer unexpected latrine purchase. Neither of the interactions are statistically significantly different from zero. We therefore treat the estimates in Table 3, Table 4, which pool across both treatment arms, as our preferred specifications.

Discussion and conclusion

This paper shows that providing finance can dramatically increase willingness to pay for improved latrines. This finding provides evidence on the previously under-studied question of whether imperfect credit markets may be in part responsible for low WTP for environmental quality in developing countries (Greenstone and Jack, 2015). Furthermore, while micro-credit to date has not been demonstrated to provide significant benefits in terms of increasing income for clients (Banerjee, 2013, Banerjee and Karlan, 2015), this paper suggests that there may be other important ways that micro-finance can enhance welfare.29

This initial investment – in this context, the purchase of a latrine – is a necessary but not sufficient condition for improving environmental conditions and health. We find that, despite the large increase in WTP for the initial investment, installation rates are low even 18–24 months after purchase. An interesting question for further research is why households exhibit preferences for expensive, elaborate superstructures and what strategies or policies might encourage completing the investment in such situations.

The dramatic increase in latrine sales in financing villages has important implications for cost-effectiveness. As with many interventions, a large share of costs are village-level fixed costs, such as the time of sales agents and their transportation costs to remote villages. Since sales agents made over four times as many sales per village meeting when loans were offered, the fixed cost of their time and transportation was amortized over many more latrines sold. Of course, offering finance carries a cost, but because VisionFund was already operating in program villages, the marginal cost of managing and collecting on loans was low. Using conservative assumptions on the costs of sales and marketing, and the marginal cost of providing finance, we calculate that offering finance can reduce program costs per latrine sold by up to 70%. For example, a direct-sales intervention, in which a team of 8 full-time sales agents travel from village to village offering latrines at USD 50 without financing, would incur operational costs (sales and marketing) of USD 19 per latrine sold. Providing financing would add to operational costs by requiring a loan officer travel to process and collect loans (note we are making the conservative assumption that an additional MFI employee would be necessary, which may not be true if the MFI already has a robust presence), but even net of this financing cost, the increase in demand through financing reduces total operational costs per latrine sold to approximately 6 USD.30 Of course, this result is muted by the low installation rate we observed – if only roughly one in three purchased latrines are installed, then the cost to the implementer per latrine installed must be scaled up appropriately.31

It is important to note that the financing in our evaluation was offered only alongside the supply of materials for latrine construction and we thus cannot extend our results directly to contexts where financing is offered separately from the provision of these materials. The relevance of this limitation depends on whether one takes the perspective of a sanitation promotion organization looking to expand coverage or that of a microfinance institution looking to expand its clientele.

Finally, this paper provides a model for learning from a “decision-focused evaluation,” i.e., one designed to answer a specific programmatic question for an implementer rather than a general academic question. However, this does come at a cost, in that we are limited in our ability to explain mechanisms underlying the large effect of credit on demand. In particular, either credit constraints or impatience could explain this reduced-form result, and further research is needed to understand the relative contributions of each. Similarly, we offer evidence on the role of peer effects on installation, but credit constraints and impatience may also drive the low installation rates.

The benefit of a “decision-focused evaluation” is that implementers and policymakers can quickly adapt their programming. Financing is now much more widely available as a routine part of iDE's latrine distribution efforts, and iDE is currently working towards implementing a one-stop shop that includes installation, a shelter, and financing for both.